Master Thesis, On the Absolute Anabelian Geometry of Hyperbolic Polycurves over Local Fields (Title, 仮)

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Contents

Introduction

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Notations and Conventions

Numbers: The notation $\mathfrak{Primes}$ will be used to denote the set of prime numbers. The notation $\mathbb{N}$ will be used to denote the set of nonnegative integers. The notation $\mathbb{Z}$ will be used to denote the additive group or ring of integers. The notation $\mathbb{Q}$ will be used to denote the field of rational numbers. The notation $\mathbb{Q}_{\geq 0}$ will be used to denote the additive monoid of nonnegative rational numbers. The notation $\mathbb{R}$ will be used to denote the field of real numbers. For each $x \in \mathbb{R}$, the notation $\lfloor x \rfloor$ will be used to denote the greatest integer $\leq$ x. If p is a prime number, then the notation $\mathbb{Q}_p$ will be used to denote the field of p-adic numbers; the notation $\mathbb{Z}_p$ will be used to denote the additive group or ring of p-adic integers; the notation $\overline{\mathbb{Q}}_p$ will be used to denote an algebraic closure of $\mathbb{Q}_p$; the notation $\mathbb{C}_p$ will be used to denote the p-adic completion of $\overline{\mathbb{Q}}_p$. We shall refer to a finite extension field of $\mathbb{Q}_p$ as a **p-adic local field**.

Monoids: Let M be a commutative monoid. In this subsection, we regard the set of positive integers $\mathbb{N}_{\geq 1}$ as a directed set via its **multiplicative** structure, i.e., $i \leq j \Leftrightarrow i | j$. For $i \in \mathbb{N}_{\geq 1}$, write $M_i \coloneqq M$. For $i, j \in \mathbb{N}_{\geq 1}$ such that $i \leq j$, write $\phi_{i,j} : M_i \to M_j$ for the homomorphism of monoids obtained by multiplication by $\frac{j}{i}$. Then we shall refer to the inductive limit of the inductive system $\{M_i, \phi_{i,j}\}$ on the directed set $\mathbb{N}_{\geq 1}$ as the **perfection** of M.

Rings and fields: Let R be a ring. Then we shall write $R^{\times}$ for the multiplicative group of units of the ring.

Let F be a perfect field, p a prime number. Then the notation $\overline{F}$ will be used to denote an algebraic closure of F; $G_F \coloneqq \mathrm{Gal}(\overline{F}/F)$.

Suppose that F is a **valuation field**, i.e., a field equipped with a valuation map. Then we shall write $\mathcal{O}_F$ for the ring of integers of F; $\mathfrak{m}_F$ for the maximal ideal of $\mathcal{O}_F$.

Topological groups: Let G be a topological group; $H \subseteq G$ a closed subgroup of G. Then we shall write G^{ab} for the abelianization of G; $Z_G(H)$ for the **centralizer** of H in G, i.e., $Z_G(H) \coloneqq \{g \in G \mid ghg^{-1} = h \text{ for all } h \in H\}$; $N_G(H)$ for the **normalizer** of H in G, i.e., $N_G(H) \coloneqq \{g \in G \mid gHg^{-1} = H\}$; and $C_G(H)$ for the **commensurator** of $H \subseteq G$, i.e.,

$$C_G(H) \coloneqq \{g \in G \mid H \cap gHg^{-1} \text{ is of finite index in } H \text{ and } gHg^{-1}\}.$$

We shall say that the closed subgroup H is **commensurably terminal** in G if $H = C_G(H)$. Let $\Sigma \subseteq \mathfrak{Primes}$ be a nonempty subset. Then we shall write G^{Σ} for the pro-Σ completion of G.

We shall write $\mathrm{Aut}(G)$ for the group of continuous automorphisms of the topological group G, $\mathrm{Inn}(G) \subseteq \mathrm{Aut}(G)$ for the subgroup of inner automorphisms of G, and $\mathrm{Out}(G) \coloneqq \mathrm{Aut}(G)/\mathrm{Inn}(G)$. Suppose that G is **center-free**. Then we have a **natural exact sequence** of groups

$$1 \to \mathrm{Inn}(G) \to \mathrm{Aut}(G) \to \mathrm{Out}(G) \to 1$$

where G is identified with $\mathrm{Inn}(G)$ via the natural isomorphism $G \cong \mathrm{Inn}(G)$ sending $g \in G$ to the inner automorphism $x \mapsto gxg^{-1}$ of G.

If J is a group, and $\rho : J \to \mathrm{Out}(G)$ is a homomorphism, then we shall denote by $G \overset{\mathrm{out}}{\rtimes} J$ the group obtained by pulling back the above exact sequence of groups via ρ. Thus, we have a natural exact sequence of groups

$$1 \to G \to G \overset{\mathrm{out}}{\rtimes} J \to J \to 1.$$

Suppose further that the topology of G admits a **countable basis** consisting of **characteristic open subgroups** of G. Then one verifies immediately that the topology of G induces a **natural topology** on the group $\mathrm{Aut}(G)$, hence on the group $\mathrm{Out}(G)$. In particular, one verifies easily that if J is a **topological group**, and $\rho : J \to \mathrm{Out}(G)$ is **continuous**, then $G \overset{\mathrm{out}}{\rtimes} J$ admits a natural **topological group** structure.

Let G_1, G_2 be topological groups. Then we shall write $\mathrm{Isom}(G_1, G_2)$ for the set of **isomorphisms** from G_1 to G_2;

$$\mathrm{OutIsom}(G_1, G_2)$$

for the set of **outer isomorphisms** from G_1 to G_2, i.e., the set of $\mathrm{Inn}(G_2)$-orbits of $\mathrm{Isom}(G_1, G_2)$ under the natural action of $\mathrm{Inn}(G_2)$ on $\mathrm{Isom}(G_1, G_2)$ by post-composition.

Schemes: Let K be a field; $K \subseteq L$ a field extension; X an algebraic variety [i.e., a separated, geometrically integral scheme of finite type] over K. Then we shall write $X_L \coloneqq X \times_K L$. Suppose that K is a valuation field. Let X be an $\mathcal{O}_K$-scheme. Then we shall write X_s for the special fiber of X [i.e., the fiber of X over the closed point of $\mathrm{Spec}\, \mathcal{O}_K$].

Let S_1, S_2 be schemes. Then we shall write $\mathrm{Isom}(S_1, S_2)$ for the set of isomorphisms of schemes between S_1 and S_2.

Log schemes: If $X^{\log}$ is a fine log scheme, then we shall write

- X for the underlying scheme of $X^{\log}$;
- $\mathcal{M}_{X^{\log}}$ for the étale sheaf of monoids on X that defines the log structure of $X^{\log}$;
- $\mathcal{C}_{X^{\log}} \coloneqq \mathcal{M}_{X^{\log}}/\mathcal{O}_X^{\times}$, which we shall refer to as the characteristic of $X^{\log}$ [cf. [MT], Definition 5.1, (i)].

Let K be a complete discrete valuation field; Y a hyperbolic curve over K; Y a compactified semistable model of Y over $\mathcal{O}_K$. Write $S \coloneqq \mathrm{Spec}\, \mathcal{O}_K$; $S^{\log}$ for the log scheme determined by the log structure associated $\mathrm{Spec}\mathcal{O}_K$; $S^{\log}$ for the log scheme determined by the log structure associated to the closed point of S. Then it follows immediately from [Hur], § 3.7, § 3.8, that the multiplicative monoid of sections of O_Y that are invertible on [the open subscheme of Y determined by] Y determines a natural log structure on Y. Denote the resulting log scheme by $Y^{\log}$. Then one verifies immediately that the natural morphism of schemes $Y \to S$ extends to a a proper, log smooth morphism $Y^{\log} \to S^{\log}$ of fine log schemes. Let y be a geometric point of Y_s. Write $\mathcal{C}^{\mathrm{pf}}_{Y^{\log},y}$ for the perfection of the stalk of the characteristic $\mathcal{C}_{Y^{\log}}$ at y. Then one verifies immediately that, if the image of y in Y_s is a smooth point that is not a cusp (respectively, a cusp; a node), then $\mathcal{M}^{\mathrm{pf}}_{Y^{\log},y} \xrightarrow{\sim} \mathbb{Q}_{\geq 0}$ (respectively, $\mathcal{M}^{\mathrm{pf}}_{Y^{\log},y} \xrightarrow{\sim} \mathbb{Q}_{\geq 0} \times \mathbb{Q}_{\geq 0}$; $\mathcal{M}^{\mathrm{pf}}_{Y^{\log},y} \xrightarrow{\sim} \mathbb{Q}_{\geq 0} \times \mathbb{Q}_{\geq 0}$).

Fundamental groups: For a connected noetherian scheme S, we shall write Π_S for the étale fundamental group of S, relative to a suitable choice of basepoint. Let $\Sigma \subseteq \mathfrak{Primes}$ be a nonempty subset; K a perfect field; X an algebraic variety over K. Then we shall write

$$\Delta_X \coloneqq \Pi_{X_{\overline{K}}}; \quad \Pi_X^{(\Sigma)} \coloneqq \Pi_X / \mathrm{Ker}(\Delta_X \twoheadrightarrow \Delta_X^{\Sigma}),$$

where $\Delta_X \to \Delta_X^{\Sigma}$ denotes the natural surjection. We shall refer to Δ_X^{Σ} (respectively, $\Pi_X^{(\Sigma)}$) as the geometric pro-Σ fundamental group (respectively, geometrically pro-Σ fundamental group) of X.

ちゃんと確認して補正する必要がある。

moduli の記述, Mgr, Mbargr の記述もする

pi1, log scheme, abs Gal group, 分岐理論

1 Preliminaries on Logarithmic Geometry of Hyperbolic Polycurves

In this section, we shall prepare some facts on the [logarithmetic] geometry of hyperbolic polycurves, which will be used as basic ingredients in the subsequent sections. In particular, we shall verify that the stable compactification of a hyperbolic polycurve admits a natural log structure, which is compatible with a certain stratification. We then discuss some general properties of log schemes whose characteristics are, in some sense, étale locally constant. Finally, we introduce the notion of (pro-ℓ) decomposition groups and inertia groups, where ℓ is a prime number which is invertible on the base scheme, associated to the strata of the stable compactification of a hyperbolic polycurve, and verify some properties of these groups.

Definition 1.1. Let S be a scheme and X a scheme over S.

(i) We shall say that X is a **hyperbolic curve** (of type (g, r)) over S if there exist

- a pair of non-negative integers (g, r);
- a scheme X^{cpt} which is smooth, proper, geometrically connected, and of relative dimension 1 over S;
- a (possibly empty) closed subscheme D of X^{cpt} which is finite étale over S

such that

- $2g - 2 + r > 0$;
- any geometric fiber of X^{cpt} over S is a smooth curve of genus g;
- the finite étale morphism $D \to S$ is of degree r;
- X is isomorphic to $X^{\mathrm{cpt}} \setminus D$ as a scheme over S.

(ii) We shall say that X is a **stable curve** (of type (g, r)) over S if there exist

- a pair of non-negative integers (g, r);
- a scheme X^{cpt} which is proper, flat, geometrically connected, and of relative dimension 1 over S;
- a (possibly empty) closed subscheme D of X^{cpt} which is finite étale over S

such that

- $2g - 2 + r > 0$;
- any geometric fiber of (X^{cpt}, D) over S is a stable curve of genus g, i.e., a curve of genus g with at worst nodal singularities and with r marked points, such that any rational component of the curve contains at least three nodes or marked points;

- the finite étale morphism $D \to S$ is of degree r;
- X is isomorphic to $X^{\mathrm{cpt}} \setminus D$ as a scheme over S.

(iii) We shall say that X is a **hyperbolic polycurve** (of relative dimension n) over S (respectively, **stable polycurve** (of relative dimension n)) if there exist

- a positive integer n;
- a factorization of the structure morphism $X \to S$ (which is not necessarily uniquely determined by X)

$$X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = S$$

such that

- for each $i = 1, \ldots, n$, the morphism $X_i \to X_{i-1}$ is a hyperbolic curve over X_{i-1} (respectively, a stable curve over X_{i-1}).

We shall refer to the above morphism $X = X_n \to X_{n-1}$ as a **parametrizing morphism** for X and refer to the above factorization as a **sequence of parametrizing morphisms** or **parametrization** for X.

Definition 1.2. Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. We shall say that $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ is a **compactified stable model** of X (with respect to its parametrization) if there exist

- open immersion $X_i \hookrightarrow \mathfrak{X}_i$ for each $i = 0, \ldots, n$;
- divisors $\mathfrak{D}_i$ on $\mathfrak{X}_i$ for each $i = 1, \ldots, n$;

such that

- open immersion $X_0 \hookrightarrow \mathfrak{X}_0$ is a (natural) immersion $\operatorname{Spec} K \hookrightarrow \operatorname{Spec} \mathcal{O}_K$;
- for each $i = 1, \ldots, n$, the morphism $(\mathfrak{X}_i, \mathfrak{D}_i) \to \mathfrak{X}_{i-1}$ is a stable curve over $\mathfrak{X}_{i-1}$;
- for each $i = 1, \ldots, n$, the pullback of the stable curve $(\mathfrak{X}_i, \mathfrak{D}_i) \to \mathfrak{X}_{i-1}$ by the open immersion $X_{i-1} \hookrightarrow \mathfrak{X}_{i-1}$ is a compactification of the hyperbolic curve $X_i \to X_{i-1}$.

Proposition 1.3 (Log structure on the stable compactification of a hyperbolic polycurve)**.** Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Then there exists an fs log structure on $\mathfrak{X}$ such that the morphism $\mathfrak{X}^{\log} \to (\operatorname{Spec} \mathcal{O}_K)^{\log}$ is log smooth, where $(\operatorname{Spec} \mathcal{O}_K)^{\log}$ is the log scheme obtained by endowing $\operatorname{Spec} \mathcal{O}_K$ with the log structure associated to the closed point of $\operatorname{Spec} \mathcal{O}_K$ — therefore, the log scheme $\mathfrak{X}^{\log}$ is log regular —

and such that the interior of $\mathfrak{X}^{\log}$ (i.e., the locus where the log structure is trivial) coincides with X. Moreover, the log structure on $\mathfrak{X}$ is uniquely determined by the above conditions.

Proof. We shall proceed by induction on the relative dimension n of X over $\operatorname{Spec} K$. The case where $n = 0$ is trivial, so we assume that $n > 0$ and that the proposition holds for any hyperbolic polycurve of relative dimension less than n.

Let $X' = X_{n-1} \to X_{n-2} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ be the hyperbolic polycurve obtained by removing the top level of the parametrization of X. By the induction hypothesis, there exists a log structure on $\mathfrak{X}' = \mathfrak{X}_{n-1} \to \mathfrak{X}_{n-2} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ such that $(\mathfrak{X}')^{\log} \to (\operatorname{Spec} \mathcal{O}_K)^{\log}$ is log smooth and such that the interior of $(\mathfrak{X}')^{\log}$ coincides with X'.

Let $\Phi \colon \mathfrak{X}' \to \overline{\mathfrak{M}}_{g_n,r_n}$ be the morphism determined by the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \to \mathfrak{X}_{n-1}$, where g_n and r_n are the genus and the number of marked points of the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \to \mathfrak{X}_{n-1}$. By the induction hypothesis the morphism Φ induces a morphism of fs log schemes $(\mathfrak{X}')^{\log} \to \overline{\mathfrak{M}}^{\log}_{g_n,r_n}$, where $\overline{\mathfrak{M}}^{\log}_{g_n,r_n}$ is endowed with the natural log structure determined by its boundary, together with the fact that Φ maps the interior of $\mathfrak{X}'^{\log}$ to the interior of $\overline{\mathfrak{M}}^{\log}_{g_n,r_n}$.

By computing the log structures involved, one verifies immediately that the underlying scheme of the fs log fiber product $(\mathfrak{X}')^{\log} \times_{\overline{\mathfrak{M}}^{\log}_{g_n,r_n}} \overline{\mathfrak{C}}^{\log}_{g_n,r_n}$ coincides with the fiber product $\mathfrak{X}' \times_{\overline{\mathfrak{M}}_{g_n,r_n}} \overline{\mathfrak{C}}_{g_n,r_n}$, which is isomorphic to $\mathfrak{X}_n$ by the definition of Φ. Thus, the scheme $\mathfrak{X}_n$ admits a natural fs log structure determined by the natural isomorphism $\mathfrak{X}_n \cong \mathfrak{X}' \times_{\overline{\mathfrak{M}}_{g_n,r_n}} \overline{\mathfrak{C}}_{g_n,r_n}$, and the resulting log scheme $\mathfrak{X}_n^{\log}$ fits into a natural cartesian diagram of fs log schemes.

Since the morphism $\overline{\mathfrak{C}}^{\log}_{g_n,r_n} \to \overline{\mathfrak{M}}^{\log}_{g_n,r_n}$ is log smooth, it follows that the morphism $\mathfrak{X}_n^{\log} \to (\mathfrak{X}')^{\log}$ is log smooth. Therefore, the morphism $\mathfrak{X}_n^{\log} \to (\operatorname{Spec} \mathcal{O}_K)^{\log}$ is also log smooth. In particular, since $(\operatorname{Spec} \mathcal{O}_K)^{\log}$ is log regular, it follows that $\mathfrak{X}_n^{\log}$ is log regular. Moreover, by the computation of the log structures involved, one verifies immediately that the interior of $\mathfrak{X}_n^{\log}$ coincides with X_n. Finally, since the log structure on a log regular fs log scheme is uniquely determined by its interior, it follows that the log structure on $\mathfrak{X}_n$ is uniquely determined by the conditions described in the proposition. This completes the proof of the proposition. □

Lemma 1.4 (Normality of the stable compactification)**.** *Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Then $\mathfrak{X}_n$ is integral and normal. Moreover, the complement of X_n in $\mathfrak{X}_n$ is purely of codimension 1.*

Proof. In light of the Proposition 1.3, the scheme $\mathfrak{X}_n$ admits a natural log structure such that $\mathfrak{X}_n^{\log}$ is log regular. In particular, the underlying scheme $\mathfrak{X}_n$ of $\mathfrak{X}_n^{\log}$ is normal and the complement of the interior of $\mathfrak{X}_n^{\log}$ (which coincides with X_n [cf. Proposition 1.3]) in $\mathfrak{X}_n$ is purely of codimension 1. Therefore, to verify Lemma 1.4, it suffices to show that $\mathfrak{X}_n$ is connected. However, this follows immediately from the elementary scheme theory and the fact that $\mathfrak{X}_n \to \operatorname{Spec} \mathcal{O}_K$ is proper and geometrically connected. This completes the proof of the lemma. $\square$

Lemma 1.5 (Uniqueness of the stable compactification of a hyperbolic polycurve)**.** Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Then any compactified stable model of X with respect to its parametrization is isomorphic to $\mathfrak{X}$ as a sequence of morphisms over $\operatorname{Spec} \mathcal{O}_K$.

Proof. We shall proceed by induction on the relative dimension n of X over $\operatorname{Spec} K$. The case where $n = 0$ is trivial, so we assume that $n > 0$ and that the lemma holds for any hyperbolic polycurve of relative dimension less than n.

Let $\mathfrak{X}' = \mathfrak{X}'_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ be another compactified stable model of X with respect to its parametrization. Since the moduli stack $\overline{\mathfrak{M}}_{g_n,r_n}$ of stable curves of genus g_n with r_n marked points is separated and Deligne-Mumford, it follows that the diagonal morphism $\overline{\mathfrak{M}}_{g_n,r_n} \to \overline{\mathfrak{M}}_{g_n,r_n} \times \overline{\mathfrak{M}}_{g_n,r_n}$ is proper and unramified; in particular, it is a finite morphism.

Let $\Phi \colon \mathfrak{X}_{n-1} \to \overline{\mathfrak{M}}_{g_n,r_n}$ (respectively, $\Phi' \colon \mathfrak{X}_{n-1} \to \overline{\mathfrak{M}}_{g_n,r_n}$) be the morphism determined by the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \to \mathfrak{X}_{n-1}$ (respectively, the stable curve $(\mathfrak{X}'_n, \mathfrak{D}'_n) \to \mathfrak{X}_{n-1}$). Since the restrictions of Φ and Φ' to X_{n-1} coincide, we obtain a commutative diagram

$$\begin{tikzcd}
X_{n-1} \arrow[r] \arrow[d] & \overline{\mathfrak{M}}_{g_n,r_n} \arrow[d, "\Delta"] \\
\mathfrak{X}_{n-1} \arrow[r, "{(\Phi, \Phi')}"] & \overline{\mathfrak{M}}_{g_n,r_n} \times \overline{\mathfrak{M}}_{g_n,r_n}.
\end{tikzcd}$$

Note that Δ is finite (and unramified) morphism since $\overline{\mathfrak{M}}_{g_n,r_n}$ is separated and Deligne-Mumford algebraic stack, and the fiber product $\mathfrak{X}_{n-1} \times_{\overline{\mathfrak{M}}_{g_n,r_n} \times \overline{\mathfrak{M}}_{g_n,r_n}} \overline{\mathfrak{M}}_{g_n,r_n}$ is just the Isom-scheme $\mathrm{Isom}_{\mathfrak{X}_{n-1}}(\mathfrak{X}_n, \mathfrak{X}'_n)$, which is representable by a scheme. Since $\mathfrak{X}_{n-1}$ is normal by Lemma 1.4, it follows that the morphism $(\Phi, \Phi') \colon \mathfrak{X}_{n-1} \to \overline{\mathfrak{M}}_{g_n,r_n} \times \overline{\mathfrak{M}}_{g_n,r_n}$ factors through Δ; in particular, we obtain the equality $\Phi = \Phi'$. Therefore, by the definition involved, we obtain an isomorphism $\mathfrak{X}_n \cong \mathfrak{X}'_n$ over $\mathfrak{X}_{n-1}$, which completes the proof of the lemma. $\square$

Definition 1.6. Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization.

(i) Let $x \in \mathfrak{X}_n$ be a point. Write x_i for the image of x in X_i for each $i = 0, \dots, n$. We shall say that x is a **stratum point** of $\mathfrak{X}_n$ if the following condition is satisfied:

   - for each $i = 1, \dots, n$, the point x_i is either a node of the stable curve $(\mathfrak{X}_i, \mathfrak{D}_i) \times_{\mathfrak{X}_{i-1}} x_{i-1} \to x_{i-1}$, a marked point of the stable curve $(\mathfrak{X}_i, \mathfrak{D}_i) \times_{\mathfrak{X}_{i-1}} x_{i-1} \to x_{i-1}$, or a generic point of an irreducible component of the stable curve $(\mathfrak{X}_i, \mathfrak{D}_i) \times_{\mathfrak{X}_{i-1}} x_{i-1} \to x_{i-1}$.

(ii) Let x be a stratum point of $\mathfrak{X}_n$. Let x_i be the image of x in $\mathfrak{X}_i$ for each $i = 0, \dots, n$. We shall refer to as **VCNGS-signature** of x the function $\mathfrak{sgn}(x) \colon \{0, \dots, n\} \to \{\mathrm{V}, \mathrm{C}, \mathrm{N}, \mathrm{G}, \mathrm{S}\}$ defined as follows:

   - for each $i = 1, \dots, n$, $\mathfrak{sgn}(x)(i) = \mathrm{V}$ if x_i is a generic point of an irreducible component of the stable curve $(\mathfrak{X}_i, \mathfrak{D}_i) \times_{\mathfrak{X}_{i-1}} x_{i-1} \to x_{i-1}$;
   - for each $i = 1, \dots, n$, $\mathfrak{sgn}(x)(i) = \mathrm{C}$ if x_i is a marked point of the stable curve $(\mathfrak{X}_i, \mathfrak{D}_i) \times_{\mathfrak{X}_{i-1}} x_{i-1} \to x_{i-1}$;
   - for each $i = 1, \dots, n$, $\mathfrak{sgn}(x)(i) = \mathrm{N}$ if x_i is a node of the stable curve $(\mathfrak{X}_i, \mathfrak{D}_i) \times_{\mathfrak{X}_{i-1}} x_{i-1} \to x_{i-1}$;
   - for $i = 0$, $\mathfrak{sgn}(x)(i) = \mathrm{G}$ if x_i is the generic point of $\operatorname{Spec} \mathcal{O}_K$;
   - for $i = 0$, $\mathfrak{sgn}(x)(i) = \mathrm{S}$ if x_i is the closed point of $\operatorname{Spec} \mathcal{O}_K$.

(iii) Let x be a stratum point of $\mathfrak{X}_n$. We shall say that the **combinatorial dimension** of x is the number of $i \in \{0, \dots, n\}$ such that $\mathfrak{sgn}(x)(i) \in \{\mathrm{V}, \mathrm{G}\}$.

Lemma 1.7 (Stratification of the stable compactification)**.** Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Then there exists a stratification of $\mathfrak{X}_n$ such that

- the strata of the stratification are irreducible;
- the set of stratum points of $\mathfrak{X}_n$ coincides with the set of generic points of the strata of the stratification.

The stratification of $\mathfrak{X}_n$ is uniquely determined by the above conditions. Moreover, for each stratum A of the stratification associated to $\mathfrak{X}_n$, the image of A in $\mathfrak{X}_i$ is one of the strata of the stratification associated to $\mathfrak{X}_i$ for each $i = 0, \dots, n-1$.

Proof. First, we shall construct a stratification of $\mathfrak{X}_n$ by induction on the relative dimension n of X over $\operatorname{Spec} K$. The case where $n = 0$ is trivial, so we assume that $n > 0$ and that the lemma holds for any hyperbolic polycurve of relative dimension less than n.

Let π be the morphism $\mathfrak{X}_n \to \mathfrak{X}_{n-1}$. For each strata A of the stratification of $\mathfrak{X}_{n-1}$ obtained by the induction hypothesis, we shall construct a decomposition of $\pi^{-1}(A)$ by locally closed subsets as follows. Let η_A be the generic point of A. Then the set of stratum points over η_A coincides with the set of nodes, marked points, and generic points of irreducible components of the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \times_{\mathfrak{X}_{n-1}} \eta_A \to \eta_A$. For the stratum points s over η_A which are nodes or marked points of the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \times_{\mathfrak{X}_{n-1}} \eta_A \to \eta_A$, we define the subscheme B_s to be the closure of s in $\pi^{-1}(A)$. For the stratum points t over η_A which are generic points of irreducible components of the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \times_{\mathfrak{X}_{n-1}} \eta_A \to \eta_A$, we define the subscheme B_t as follows: let $s_1, \ldots, s_m$ be the nodes and marked points of the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \times_{\mathfrak{X}_{n-1}} \eta_A \to \eta_A$ which are contained in the irreducible component corresponding to t, and let $B_{s_1}, \ldots, B_{s_m}$ be the strata defined above corresponding to $s_1, \ldots, s_m$. Then we define B_t to be the complement of $\bigcup_{i=1}^m B_{s_i}$ in the closure of t in $\pi^{-1}(A)$.

Let $t_1, \ldots, t_l$ be the stratum points of $\mathfrak{X}_n$ over η_A which are generic points of irreducible components of the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \times_{\mathfrak{X}_{n-1}} \eta_A \to \eta_A$, and let $s_1, \ldots, s_m$ be the stratum points of $\mathfrak{X}_n$ over η_A which are nodes or marked points of the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \times_{\mathfrak{X}_{n-1}} \eta_A \to \eta_A$. Note that it follows immediately from the construction and induction hypothesis that the strata $B_{t_1}, \ldots, B_{t_l}, B_{s_1}, \ldots, B_{s_m}$ are irreducible, locally closed in $\mathfrak{X}_n$, and mutually disjoint. Moreover, by the going-down theorem, every point of $\pi^{-1}(A)$ has a generalization which is a stratum point over η_A. Therefore, $B_{t_1}, \ldots, B_{t_l}, B_{s_1}, \ldots, B_{s_m}$ constitute a decomposition of $\pi^{-1}(A)$ by locally closed subsets. Since every stratum point of $\mathfrak{X}_n$ is over a stratum point of $\mathfrak{X}_{n-1}$, it follows that the decompositions obtained by performing the above construction for all strata of $\mathfrak{X}_{n-1}$ constitute a decomposition of $\mathfrak{X}_n$ such that the set of stratum points of $\mathfrak{X}_n$ coincides with the set of generic points of the strata of the decomposition.

Finally, since the closure of a stratum point x_n of $\mathfrak{X}_n$ is either a finite étale cover over the closure of the image x_{n-1} or an underlying scheme of a stable curve over the closure of the image x_{n-1} of x_n in $\mathfrak{X}_{n-1}$, it follows from the induction hypothesis and the various definitons involved that the above decomposition of $\mathfrak{X}_n$ is indeed a stratification.

The remaining assertions of the lemma follow immediately from the construction and the definitions involved. This completes the proof of the lemma. □

Lemma 1.8 (Dimension of strata of the stable compactification)**.** Let K be a complete discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n. Then for each stratum A of $\mathfrak{A}$, the dimension of the closure of A in $\mathfrak{X}_n$ coincides with the combinatorial dimension of the stratum points corresponding to the stratum A.

Proof. By noting that $\mathcal{O}_K$ is excellent (in particular, universally catenary), it follows immediately from the elementary scheme theory and the various definitions involved. □

Lemma 1.9 (Strata of codimension 0 and 1 of the stable compactification)**.** Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 =$

$\operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n.

(i) Let x be a stratum point of $\mathfrak{X}_n$, and let d_x be the combinatorial dimension of x. Then the following conditions are equivalent:

  (a) $d_x = n + 1$;

  (b) the stratum point x is a generic point of $\mathfrak{X}_n$.

  Moreover, if the above conditions are satisfied, the corresponding stratum coincides with X_n as a subscheme of $\mathfrak{X}_n$.

(ii) Let x be a stratum point of $\mathfrak{X}_n$, and let d_x be the combinatorial dimension of x. Then the following conditions are equivalent:

  (a) $d_x = n$;

  (b) the stratum point x is a generic point of an irreducible component of the divisor at infinity of $\mathfrak{X}_n$, i.e., a generic point of an irreducible component of the closed subspace $\mathfrak{X}_n \setminus X_n$ of $\mathfrak{X}_n$.

  Moreover, for every irreducible component D of the divisor at infinity of $\mathfrak{X}_n$, the generic point of D is a stratum point of $\mathfrak{X}_n$.

Proof. It follows immediately from the definitions involved. □

Definition 1.10. Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n.

(i) We shall say that an irreducible component D of the divisor at infinity of $\mathfrak{X}_n$, i.e., an irreducible component of the closed subspace $\mathfrak{X}_n \setminus X_n$ of $\mathfrak{X}_n$, is a **divisor of cusp-type** if the following condition is satisfied:

  - there exists an element $i \in \{1, \ldots, n\}$ such that $\mathfrak{sgn}(\eta_D)(i) = \mathrm{C}$, where η_D is the generic point of D.

  Moreover, we shall say that the number of $i \in \{1, \ldots, n\}$ such that $\mathfrak{sgn}(\eta_D)(i) = \mathrm{C}$ is the **C-level** of D.

(ii) We shall say that an irreducible component D of the divisor at infinity of $\mathfrak{X}_n$, i.e., an irreducible component of the closed subspace $\mathfrak{X}_n \setminus X_n$ of $\mathfrak{X}_n$, is a **divisor of vertex-type** if the following condition is satisfied:

  - for every $i \in \{1, \ldots, n\}$, $\mathfrak{sgn}(\eta_D)(i) = \mathrm{V}$, where η_D is the generic point of D.

Lemma 1.11 (Strata and Divisors of Infinity of the stable compactification)**.** Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n. Then for each stratum A of $\mathfrak{A}$ and each irreducible component of the divisor at infinity D of $\mathfrak{X}_n$, i.e., each irreducible component of the closed subspace $\mathfrak{X}_n \setminus X_n$ of $\mathfrak{X}_n$, A is either contained in D or disjoint from D.

Proof. By the construction of the stratification of $\mathfrak{X}_n$, which is given in the proof of Lemma 1.7, it follows immediately that the divisor D is a disjoint union of strata of the stratification of $\mathfrak{X}_n$. Therefore, for each stratum A of the stratification of $\mathfrak{X}_n$, the intersection $A \cap D$ is either empty or coincides with A. This completes the proof of the lemma. $\square$

Lemma 1.12 (Strata and Divisors of Infinity of the moduli stack of stable curves)**.** Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_{n-1}$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_{n-1}, and let $\Phi\colon \mathfrak{X}_{n-1} \to \overline{\mathfrak{M}}_{g_n,r_n}$ be the morphism determined by the stable curve $(\mathfrak{X}_n, \mathfrak{D}_n) \to \mathfrak{X}_{n-1}$. Then for each stratum A of $\mathfrak{A}$ and each irreducible component of the divisor at infinity D of the moduli stack $\overline{\mathfrak{M}}_{g_n,r_n}$, i.e., each irreducible component of the boundary $\overline{\mathfrak{M}}_{g_n,r_n} \setminus \mathfrak{M}_{g_n,r_n}$, the intersection $\Phi^{-1}(D) \cap A$ is either empty or coincides with A.

Proof. By noting that $\overline{\mathfrak{M}}_{g_n,r_n}$ is a smooth Deligne-Mumford algebraic stack over $\operatorname{Spec} \mathbb{Z}$, it follows that for each divisor of $\overline{\mathfrak{M}}_{g_n,r_n}$, the pullback of the divisor to $\mathfrak{X}_{n-1}$ is a Cartier divisor. Thus, the assertion follows formally from the Lemma 1.11. This completes the proof of the lemma. $\square$

Definition 1.13. Let $X^{\log}$ be a locally Noetherian fs log scheme.

(i) We shall say that $X^{\log}$ is **log-constant**, if there exists an fs monoid P and the chart $\underline{P} \to \mathcal{M}_{X^{\log}}$ of the log structure $\mathcal{M}_{X^{\log}}$ on X such that the induced morphism $\underline{P} \to \mathcal{C}_{X^{\log}}$ (where $\underline{P}$ is the constant sheaf associated to P) is an isomorphism. We shall refer to such a monoid P as a **type** of $X^{\log}$, and we shall say that $X^{\log}$ is of type P.

(ii) We shall say that $X^{\log}$ is **étale locally log-constant** at $x \in X$, if there exists an étale neighborhood U of x such that the restriction of $X^{\log}$ to U is log-constant. We shall say that $X^{\log}$ is **étale locally log-constant**, if $X^{\log}$ is étale locally log-constant at every point of X. For each connected component of X, we shall say that a monoid P is a **type** of $X^{\log}$ at the connected component if there exists an étale covering U of the connected component such that the restriction of $X^{\log}$ to U is log-constant of type P. If X is connected, we shall simply say that P is a type of $X^{\log}$.

(iii) We shall say that $X^{\log}$ is **characteristic-constant**, if the étale sheaf of characteristic $\mathcal{C}_{X^{\log}}$ of $X^{\log}$ is a constant sheaf. Let P be a monoid. Then we shall refer to an isomorphism $\underline{P} \cong \mathcal{C}_{X^{\log}}$ as a **trivialization of characteristic** of $X^{\log}$ (of type P).

(iv) Let $X^{\log}$ be an étale locally log-constant fs log scheme. Let n be an integer. Then we shall say that $X^{\log}$ is of **log-rank** n if there exists an fs monoid P such that $X^{\log}$ is of type P and the rank of P^{gp} is equal to n.

Remark 1.14. The above definitions are an immediate generalization of the notion of "a morphism of type $\mathbb{N}^{\oplus n}$", which is introduced in [Hoshi1].

Remark 1.15. We can easily generalize the above definitions to the relative situation, i.e., for a morphism of fs log schemes $f \colon X^{\log} \to Y^{\log}$ whose underlying morphism of schemes is an isomorphism, we can define the notions of f being log-constant, étale locally log-constant, or characteristic-constant by replacing $\mathcal{C}_{X^{\log}}$ with the étale sheaf of relative characteristic $\mathcal{C}_{X^{\log}/Y^{\log}}$ of f in the above definitions. However, we shall not need such a generalization in this paper, and therefore we do not give a precise definition of these notions.

Lemma 1.16 (Characterization of étale locally log-constant fs log schemes)**.** Let $X^{\log}$ be an fs log scheme. Then the following conditions are equivalent:

(i) $X^{\log}$ is étale locally log-constant;

(ii) the étale sheaf of characteristic $\mathcal{C}_{X^{\log}}$ of $X^{\log}$ is étale locally constant.

Proof. It follows immediately from the definitions involved. □

Lemma 1.17 (Correspondence between characteristic-constant log schemes and certain homomorphisms to the Picard group)**.** Let X be a reduced scheme and P be an fs sharp monoid. Write $\mathbf{M}_X$ for the set of isomorphism classes of pairs $(\mathcal{M}, \alpha)$ such that

- $\mathcal{M}$ is an fs log structure on X;
- Let $X^{\log}$ be the fs log scheme whose underlying scheme is X and whose log structure is $\mathcal{M}$. Then $X^{\log}$ is characteristic-constant and of type P;
- $\alpha \colon \underline{P} \to \mathcal{C}_{(X,\mathcal{M})}$ is an isomorphism.

Write $\iota_X \colon \mathbf{M}_X \to \mathrm{Hom}(P, \mathrm{Pic}(X)) = \mathrm{Hom}(P^{\mathrm{gp}}, \mathrm{Pic}(X))$ for the map defined as follows:

- For each element $(\mathcal{M}, \alpha)$ of $\mathbf{M}_X$, write $X^{\log}$ for the fs log scheme whose underlying scheme is X and whose log structure is $\mathcal{M}$. Then we define $\iota_X(\mathcal{M}, \alpha)$ to be the homomorphism $P \to \mathrm{Pic}(X)$ defined as follows: for each element p of P, we define $\iota_X(\mathcal{M}, \alpha)(p)$ to be the isomorphism class of the invertible sheaf $(\alpha \circ \mathrm{pr})^{-1}(p)$ on X, where $\mathrm{pr} \colon \mathcal{M}_{X^{\log}} \to \mathcal{C}_{X^{\log}}$ is the natural projection.

Then the map ι_X is a bijection.

Proof. First, we shall construct the (inverse) map $\kappa_X : \mathrm{Hom}(P, \mathrm{Pic}(X)) \to \mathbf{M}_X$ of ι_X. Let $f\colon P \to \mathrm{Pic}(X)$ be a homomorphism. Write $f^{\mathrm{gp}}\colon P^{\mathrm{gp}} \to \mathrm{Pic}(X)$ for the homomorphism induced by f on the groupifications. Temporarily, we take an isomorphism $v\colon \mathbb{Z}^{\oplus n} \to P^{\mathrm{gp}}$ of abelian groups, where n is the rank of P^{gp}. For $1 \leq i \leq n$, we write $\mathbf{e}_i$ for the i-th standard basis vector of $\mathbb{Z}^{\oplus n}$. Write $\mathcal{L}_i$ for the invertible sheaf on X corresponding to $f^{\mathrm{gp}}(v(\mathbf{e}_i))$. Then we define an extension $0 \to \mathcal{O}_X^\times \to \mathcal{M} \to \underline{P} \to 0$ of sheaves on X as follows: we define $\mathcal{M}$ to be the coproduct of $\mathcal{L}_1^{\otimes a_1} \otimes \cdots \otimes \mathcal{L}_n^{\otimes a_n}$ for all $a_1, \ldots, a_n \in \mathbb{Z}$ such that $v(a_1\mathbf{e}_1 + \cdots + a_n\mathbf{e}_n) \in P$ as a sheaf of sets, and we define the monoid structure on $\mathcal{M}$ by the natural (various) isomorphisms. This construction yields an bijective correspondence between the set of extensions of $\underline{P}$ by $\mathcal{O}_X^\times$ and the set of homomorphisms from P to $\mathrm{Pic}(X)$. Next, we shall construct a log structure $\mathcal{M} \to \mathcal{O}_X$ on X. Let s be a local section of $\mathcal{M}$, and let p be an element of P such that the image of s in $\underline{P}$ coincides with p. Then we define the image of s of the morphism $\mathcal{M} \to \mathcal{O}_X$ to be s $(\in \mathcal{O}_X^\times)$ if $p = 0$ and to be 0 if $p \neq 0$. This construction yields a log structure on X, and the resulting fs log scheme is characteristic-constant and of type P. Thus, we have constructed the map $\kappa_X : \mathrm{Hom}(P, \mathrm{Pic}(X)) \to \mathbf{M}_X$.

Finally, we verify that ι_X and κ_X are inverse to each other. For each element $(\mathcal{M}, \alpha)$ of $\mathbf{M}_X$, we have an extension $0 \to \mathcal{O}_X^\times \to \mathcal{M} \to \underline{P} \to 0$ of sheaves on X, and the corresponding homomorphism $P \to \mathrm{Pic}(X)$. To verify the assertion that ι_X and κ_X are inverse to each other, it suffices to verify that the log structure $\mathcal{M} \to \mathcal{O}_X$ satisfies the following condition: for each local section s of $\mathcal{M}$ and each element p of P such that the image of s in $\underline{P}$ coincides with p, the image of s by the morphism $\mathcal{M} \to \mathcal{O}_X$ coincides with s if $p = 0$ and with 0 if $p \neq 0$. The case $p = 0$ is trivial, so we assume that $p \neq 0$. Since the condition is local, we can check it on a smaller open subset.

Let U be an étale open subset of X such that s is defined, and the (local) chart $\varphi\colon \underline{P}_U \to \mathcal{M}|_U$ of the log structure $\mathcal{M}$ on X exists. Then s can be written as $\varphi(p) \cdot u$ for some $u \in \mathcal{O}_X^\times(U)$. Since the image of $\varphi(p)$ in $\mathcal{C}$ is nonzero everywhere on U, where $\mathcal{C}$ is the étale sheaf of characteristic of $\mathcal{M}$, it follows that the image of $\varphi(p)$ by the morphism $\mathcal{M} \to \mathcal{O}_X$ is not invertible at every point of U. Therefore, since X is reduced, the image of $\varphi(p)$ by the morphism $\mathcal{M} \to \mathcal{O}_X$ is 0. This completes the verification that ι_X and κ_X are inverse to each other, and therefore completes the proof of the lemma. $\square$

Definition 1.18. Let X be a reduced scheme and P be an fs sharp monoid. Let $(X^{\log}, \alpha)$ be a pair consisting of an fs log scheme $X^{\log}$ whose underlying scheme is X and an isomorphism $\alpha\colon \underline{P} \to \mathcal{C}_{X^{\log}}$. Let $f\colon P \to \mathrm{Pic}(X)$ be the homomorphism corresponding to $(X^{\log}, \alpha)$ via the bijection ι_X in Lemma 1.17. We shall say that the pair $(X^{\log}, \alpha)$ is associated to the homomorphism f.

Lemma 1.19 (Description of the log structure of the stable compactification)**.** Let K be a discrete valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n. Let A be a stratum of $\mathfrak{A}$. Then the following assertions hold:

(i) Write $A^{\log}$ for the fs log scheme whose underlying scheme is A, endowed with the reduced scheme structure, and whose log structure is the pullback of the log structure on $\mathfrak{X}_n^{\log}$. Then $A^{\log}$ is étale locally log-constant.

(ii) Let η_A be the generic point of A. Let π be the morphism $\mathfrak{X}_n \to \mathfrak{X}_{n-1}$, and let $\overline{A}$ be the image of A in $\mathfrak{X}_{n-1}$ by π. Let $\eta_{\overline{A}}$ be the generic point of $\overline{A}$. We note that η_A and $\eta_{\overline{A}}$ are stratum points of $\mathfrak{X}_n$ and $\mathfrak{X}_{n-1}$, respectively. Define $A^{\log}$, $\overline{A}^{\log}$ for the fs log schemes as in (i) above. Then the following assertions hold:

  (a) If $\mathfrak{sgn}(\eta_A)(n) = \mathrm{V}$, then the morphism $A^{\log} \to \overline{A}^{\log}$ induced by π is strict, i.e., the type of $A^{\log}$ coincides with the type of $\overline{A}^{\log}$.

  (b) If $\mathfrak{sgn}(\eta_A)(n) = \mathrm{C}$, then there exists a natural isomorphism $\mathcal{C}_{A^{\log}} \cong \mathcal{C}_{\overline{A}^{\log}}|_A \oplus \underline{\mathbb{N}}$ of étale sheaves on A, where $\underline{\mathbb{N}}$ is the constant sheaf associated to $\mathbb{N}$.

  (c) Let P be the type of $A^{\log}$ and $\overline{P}$ be the type of $\overline{A}^{\log}$. If $\mathfrak{sgn}(\eta_A)(n) = \mathrm{N}$, then there exists a pushout diagram of monoids:

$$\begin{array}{ccc} \mathbb{N} & \longrightarrow & \overline{P} \\ {\scriptstyle\Delta}\downarrow & & \downarrow \\ \mathbb{N}^{\oplus 2} & \longrightarrow & P. \end{array}$$

(iii) Let η_A be the generic point of A, and let d_A be the combinatorial dimension of η_A. Then the log rank of $A^{\log}$ is equal to $n + 1 - d_A$.

Proof. In light of Lemma 1.12, it follows immediately from the description of the (log) structure of the stable curves (cf. [KatoF]) and the various definitions involved. $\square$

Lemma 1.20 (Finiteness of automorphism groups of fs sharp monoids)**.** Let P be an fs sharp monoid. Then the automorphism group $\mathrm{Aut}(P)$ of P is finite.

Proof. It follows from the well-known fact that the set of irreducible elements of P, i.e., the set of nonzero elements of P which cannot be written as a sum of two nonzero elements of P, is finite and generates P as a monoid, that the automorphism group $\mathrm{Aut}(P)$ of P is finite. This completes the proof of the lemma. $\square$

Lemma 1.21 (Representability of the functor of trivializations of characteristic of étale locally log-constant fs log schemes)**.** Let $X^{\log}$ be a locally Noetherian, connected fs log scheme which is étale locally log-constant. Let P be a monoid which is a type of $X^{\log}$. Define a functor $F_{X,P}$ from the category of schemes Sch to the category of sets Set as follows: for each scheme S, we define $F_{X,P}(S)$ to be the set of pairs (f, α) such that

- the morphism of schemes $f\colon S \to X$;
- the isomorphism $\alpha\colon \underline{P} \to f^*\mathcal{C}_{X^{\log}}$.

Then the functor $F_{X,P}$ is representable by a scheme, and the (canonical) morphism $F_{X,P} \to X$ is a finite étale covering.

Proof. Take an étale covering U of X such that the pullback of $\mathcal{C}_{X^{\log}}$ is a constant sheaf. Write F_1 for the scheme

$$\coprod_{V \subset U\,:\,\text{conn. comp.}} \ \coprod_{f \in \mathrm{Hom}(\underline{P}_V, \mathcal{C}_{X^{\log}}|_V)} V.$$

In light of the general theory of descent, and since $\mathrm{Aut}(P)$ is finite by Lemma 1.20, it follows that the scheme F_1 naturally descends to a scheme F_0 which is finite étale over X, and F_0 represents the functor $F_{X,P}$. This completes the proof of the lemma. □

Theorem 1.22 (Exact sequence of log fundamental groups of fs log schemes which are étale locally log-constant)**.** Let $X^{\log}$ be a connected, reduced fs log scheme which is étale locally log-constant. Write X for the underlying scheme of $X^{\log}$, or the log scheme obtained by endowing X with the trivial log structure. Let x be a geometric point of X, and let $\tilde{x}$ be a log geometric point of $X^{\log}$ lying over x. Then there exists an exact sequence of profinite groups

$$\pi_1(X^{\log} \times_X x, \tilde{x}) \to \pi_1(X^{\log}, \tilde{x}) \to \pi_1(X, x) \to 1.$$

Proof. First, we observe that we may assume without loss of generality that X is characteristic-constant. Indeed, by Lemma 1.21, there exists a connected finite étale covering Y of X such that the pullback of $\mathcal{C}_{X^{\log}}$ to Y is a constant sheaf. Write $Y^{\log}$ for the fs log scheme whose underlying scheme is Y and whose log structure is the pullback of the log structure on $X^{\log}$. Then $Y^{\log}$ is characteristic-constant. Now we observe that there exists a pullback diagram of profinite groups:

$$\begin{tikzcd}
\pi_1(Y^{\log} \times_Y y, \tilde{y}) \arrow[r] \arrow[d] & \pi_1(Y^{\log}, \tilde{y}) \arrow[d] \\
\pi_1(X^{\log} \times_X x, \tilde{x}) \arrow[r] & \pi_1(X^{\log}, \tilde{x}),
\end{tikzcd}$$

for some geometric point y of Y lying over x and some log geometric point $\tilde{y}$ of $Y^{\log}$ lying over y. Thus, if we verify the exactness of the sequence in the assertion for $Y^{\log}$, then it follows that the sequence in the assertion is also exact for $X^{\log}$. Therefore, we may assume without loss of generality that X is characteristic-constant.

Next, we observe that we may assume without loss of generality that $\mathbb{N}^{\oplus n}$ is the type of $X^{\log}$ for some integer n. Indeed, by the elementary monoid theory, there exists an injection $P \to \mathbb{N}^{\oplus n}$ of monoids for some integer n, which induces an isomorphism $P^{\mathrm{gp}} \cong (\mathbb{N}^{\oplus n})^{\mathrm{gp}}$. By the description of characteristic-constant log schemes, stated in Lemma 1.17, it follows that there exists a morphism $X'^{\log} \to X^{\log}$ of fs log schemes such that the underlying morphism of schemes is an isomorphism and induced morphism on the characteristic is isomorphic (as a morphism of monoid sheaves) to the injection $\underline{P} \to \underline{\mathbb{N}^{\oplus n}}$.

We now proceed to show that the (natural) morphism of Galois categories $\mathrm{F\acute{e}t}(X^{\log}) \to \mathrm{F\acute{e}t}(X'^{\log})$ is an equivalence. By the descent theory of log schemes (cf. [Vidal, Proposition 4.4], [Hoshi1, Proposition B.8]), it suffices to verify the case where X is the spectrum of a strictly henselian local ring. However, the assertion follows immediately from the description of the két covering of a strictly henselian local fs log scheme (cf. [Hoshi1, Proposition B.2]).

Finally, we verify the exactness of the sequence in the assertion for $X^{\log}$ such that $\mathbb{N}^{\oplus n}$ is the type of $X^{\log}$ for some integer n. However, it follows immediately from [Hoshi1, Corollary 2]. $\square$

Lemma 1.23 (Slimness of the extension group of slim groups)**.** Let $1 \to N \to G \to Q \to 1$ be an exact sequence of profinite groups. Assume that N and Q are slim. Then G is also slim.

Proof. Let H be an open subgroup of G. Let $g \in G$ be an element of centralizer of H. Then the image of g in Q is an element of the centralizer of the image of H in Q, and therefore the image of g in Q is trivial. Thus, g is an element of N. Since g is an element of the centralizer of $H \cap N$ in N, together with the assumption that N is slim, it follows that g is trivial. This completes the proof of the lemma. $\square$

Definition 1.24. Let K be a field, and let $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ be a hyperbolic polycurve over K. Let ℓ be a prime number. We shall say that X is **geometrically pro-ℓ-exact** if for each $0 \leq r < s < t \leq n$, then the sequence of profinite groups

$$1 \to \Delta^{\ell}_{X_t/X_s} \to \Delta^{\ell}_{X_t/X_r} \to \Delta^{\ell}_{X_s/X_r} \to 1$$

is exact, where $\Delta^{\ell}_{X_j/X_i}$ $(0 \leq i < j \leq n)$ is the maximal pro-ℓ quotient of the kernel of the morphism $\Delta_{X_j} \to \Delta_{X_i}$.

Lemma 1.25 (Slimness of the geometric pro-ℓ fundamental group of a geometrically pro-ℓ-exact hyperbolic polycurve)**.** Let K be a field, and let $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ be a hyperbolic polycurve over K. Assume that X is geometrically pro-ℓ-exact. Then the geometric pro-ℓ fundamental group Δ^{ℓ}_X of X is slim.

Proof. By Lemma 1.23 and induction on the (relative) dimension of the polycurve, and since the geometric pro-ℓ fundamental group of a hyperbolic curve is slim, the assertion follows immediately. $\square$

Lemma 1.26 (Tameness of the outer representation of a geometrically pro-ℓ-exact hyperbolic polycurve which admits a compactified stable model)**.** Let K be a complete discrete valuation field, and let $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ be a hyperbolic polycurve over K. Let ℓ be a prime number which is invertible in $\mathcal{O}_K$. Assume X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization, and that X is geometrically pro-ℓ-exact. Then the outer representation $G_K \to \operatorname{Out}(\Delta_X^{\ell})$, which is induced by the exact sequence $1 \to \Delta_X^{\ell} \to \Pi_X^{(\ell)} \to G_K \to 1$, factors through the profinite group G_K/P_K.

Proof. Let s be the closed point of $\operatorname{Spec} \mathcal{O}_K$, and let $\overline{s}^{\log}$ be a log geometric point of $\operatorname{Spec} \mathcal{O}_K^{\log}$ lying over s. Consider the commutative diagram of homomorphisms of profinite groups:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \Delta_X & \longrightarrow & \Pi_X & \longrightarrow & \pi_1(\operatorname{Spec} K) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & \pi_1(\mathfrak{X}^{\log,\mathrm{geom}}) & \to & \pi_1(\mathfrak{X}^{\log}) & \to & \pi_1(\operatorname{Spec} \mathcal{O}_K^{\log}) & \longrightarrow & 1, \end{array}$$

where $\mathfrak{X}^{\log,\mathrm{geom}}$ is the log geometric fiber of $\mathfrak{X}^{\log} \to \operatorname{Spec} \mathcal{O}_K^{\log}$ at $\overline{s}^{\log}$. We note that the profinite group $\pi_1(\operatorname{Spec} \mathcal{O}_K^{\log})$ is (naturally) isomorphic to the quotient G_K/P_K of G_K by the wild inertia subgroup P_K.

By [Kisin, Theorem 1.4], the morphism $\Delta_X^{\ell} \to \pi_1(\mathfrak{X}^{\log,\mathrm{geom}})^{\ell}$, which is obtained by taking the maximal pro-ℓ quotients of the morphism $\Delta_X \to \pi_1(\mathfrak{X}^{\log,\mathrm{geom}})$, is an isomorphism. Now, by chasing the diagram above, the assertion in the lemma follows immediately. $\square$

Definition 1.27. Let K be a complete discrete valuation field. We shall refer to the profinite group $(G_K/P_K)/(I_K/P_K)^{(\ell')}$ as the **ℓ-tame absolute Galois group** of K, where $(I_K/P_K)^{(\ell')}$ is the maximal prime-to-ℓ quotient of I_K/P_K. We shall write bsK for the ℓ-tame absolute Galois group of K.

Definition 1.28. Let K be a complete discrete valuation field, and let $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ be a hyperbolic polycurve over K. Let ℓ be a prime number which is invertible in $\mathcal{O}_K$. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization, and geometrically pro-ℓ-exact.

(i) Let $G_K \to \operatorname{Out}(\Delta_X^{\ell})$ be the outer representation induced by the exact sequence $1 \to \Delta_X^{\ell} \to \Pi_X^{(\ell)} \to G_K \to 1$. By Lemma 1.26, the outer representation $G_K \to \operatorname{Out}(\Delta_X^{\ell})$ factors through the profinite group G_K/P_K, where P_K is the wild inertia subgroup of G_K. Since Δ_X^ℓ is slim (cf. Lemma 1.25), we obtain a exact sequence of profinite groups

$$1 \to \Delta_X^\ell \to \Pi_{X,\mathrm{tame}}^{(\ell)} \to G_K/P_K \to 1,$$

where $\Pi_{X,\mathrm{tame}}^{(\ell)}$ is the pullback of profinite groups $G_K/P_K \times_{\mathrm{Out}(\Delta_X^\ell)} \mathrm{Aut}(\Delta_X^\ell)$. We shall refer to $\Pi_{X,\mathrm{tame}}^{(\ell)}$ as the **tamely reduced geometrically pro-ℓ fundamental group** of X.

(ii) We shall refer to the pullback of the profinite group $\Pi_{X,\mathrm{tame}}^{(\ell)} \times_{G_K/P_K} I_K/P_K$ as the **tamely reduced inertia geometrically pro-ℓ fundamental group** of X. We shall write $\Pi_{X,\mathrm{t.iner.}}^{(\ell)}$ for the tamely reduced inertia geometrically pro-ℓ fundamental group of X.

(iii) We shall say that X is **inertia pro-ℓ-exact** if for each $0 \leq i \leq n$, the outer representation $I_K/P_K \to \mathrm{Out}(\Delta_{X_i}^\ell)$, which is induced by the exact sequence

$$1 \to \Delta_{X_i}^\ell \to \Pi_{X_i,\mathrm{t.iner.}}^{(\ell)} \to I_K/P_K \to 1$$

factors through the maximal pro-ℓ quotient $(I_K/P_K)^\ell$ of I_K/P_K. (Here, we recall that the condition inertia pro-ℓ-exactness implies (by definition, as a tautology) the condition geometrically pro-ℓ-exactness.)

(iv) Assume that X is inertia pro-ℓ-exact. Now we observe that the outer representation $G_K/P_K \to \mathrm{Out}(\Delta_X^\ell)$, which is induced by the exact sequence in (i) above, factors through the ℓ-tame absolute Galois group $G_{K,\ell\text{-tame}}$ of K. Thus, we obtain an exact sequence of profinite groups

$$1 \to \Delta_X^\ell \to \Pi_{X,\ell\text{-tame}}^{(\ell)} \to G_{K,\ell\text{-tame}} \to 1,$$

where $\Pi_{X,\ell\text{-tame}}^{(\ell)}$ is the pullback of profinite groups $G_{K,\ell\text{-tame}} \times_{\mathrm{Out}(\Delta_X^\ell)} \mathrm{Aut}(\Delta_X^\ell)$. We shall refer to $\Pi_{X,\ell\text{-tame}}^{(\ell)}$ as the **ℓ-tamely reduced geometrically pro-ℓ fundamental group** of X. We abbreviate the term "ℓ-tamely reduced geometrically pro-ℓ fundamental group" to "**ℓ-TR geometrically pro-ℓ fundamental group**".

(v) Assume that X is inertia pro-ℓ-exact. We shall refer to the pullback of the profinite group $\Pi_{X,\ell\text{-tame}}^{(\ell)} \times_{G_{K,\ell\text{-tame}}} (I_K/P_K)^\ell$ as the **inertia pro-ℓ fundamental group** of X. We shall write $\Pi_{X,\mathrm{iner.}}^\ell$ for the inertia pro-ℓ fundamental group of X.

Definition 1.29. Let K be a discrete valuation field, and let $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \mathrm{Spec}\, K$ be a hyperbolic polycurve over K. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \mathrm{Spec}\, \mathcal{O}_K$ with respect to its parametrization. We shall say that X is **graphically inert** if for each stratum point η of $\mathfrak{X}_i$ $(0 \leq i \leq n-1)$, the action of the Galois group $\pi_1(\eta, \overline{\eta})$ on the dual graph of the fiber of $\mathfrak{X}_{i+1} \to \mathfrak{X}_i$ at η is trivial, where $\overline{\eta}$ is a geometric point of $\mathfrak{X}_i$ lying over η.

Remark 1.30. For a hyperbolic polycurve $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ over a field K, X is graphically inert if and only if for each stratum point η of X_i $(0 \leq i \leq n-1)$, the dual graph of the hyperbolic curve $X_{i+1} \times_{X_i} \eta$ over the field $\kappa(\eta)$ is split. Moreover, this condition is equivalent to the condition that for every stratum A of $\mathfrak{X}_i$ $(1 \leq i \leq n)$, letting $\overline{A}$ be the image of A in $\mathfrak{X}_{i-1}$, the induced morphism $A \to \overline{A}$ is geometrically connected. Thus, if X is graphically inert and A is a stratum of $\mathfrak{X}_i$ $(1 \leq i \leq n)$ such that $\mathfrak{sgn}(\eta_A)(i) = \mathrm{C}$ (respectively, $\mathfrak{sgn}(\eta_A)(i) = \mathrm{N}$), where η_A is the generic point of A, then the morphism $A \to \overline{A}$ is isomorphism (respectively, isomorphism).

Definition 1.31. Let K be a complete discrete valuation field, $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Let K^{ur} be the maximal unramified extension of K, let k be the residue field of K, and let $\overline{k}$ be an algebraic closure of k. Let ℓ be a prime number which is invertible in $\mathcal{O}_K$. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization, and that X is inertia pro-ℓ-exact and graphically inert. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n. Let A be a stratum of $\mathfrak{A}$. Let η_A be the generic point of A. Let $a^{\log}$ be a log geometric point of $A^{\log}$ lying over a geometric point a of A.

(i) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{G}$. Then we shall refer to the maximal pro-ℓ quotient of the kernel of the morphism $\pi_1(A^{\log}, a^{\log}) \to G_K$ (respectively, $\pi_1(A, a) \to G_K$) as the **geometric pro-ℓ decomposition group** of A (respectively, **geometric pro-ℓ base-decomposition group** of A), and we shall write $D^{\ell}_{A,\mathrm{geom.}}$ (respectively, Δ^{ℓ}_{A}) for the geometric pro-ℓ decomposition group of A (respectively, the geometric pro-ℓ base-decomposition group of A).

(ii) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{G}$. Then we shall refer to the maximal pro-ℓ quotient of the kernel of the morphism $\pi_1(A^{\log}, a^{\log}) \to \mathrm{Gal}(K^{\mathrm{ur}}/K)$ (respectively, $\pi_1(A, a) \to \mathrm{Gal}(K^{\mathrm{ur}}/K)$) as the **inertia pro-$\ell$ decomposition group** of A (respectively, **inertia pro-ℓ base-decomposition group** of A), and we shall write $D^{\ell}_{A,\mathrm{iner.}}$ (respectively, $D^{\ell}_{A,\mathrm{b.iner.}}$) for the inertia pro-$\ell$ decomposition group of A (respectively, the inertia pro-ℓ base-decomposition group of A). Here, we note that the Galois group $\mathrm{Gal}(K^{\mathrm{ur}}/K)$ is naturally isomorphic to the absolute Galois group G_k of the residue field k of K.

(iii) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{G}$. Then we obtain the exact sequence of profinite groups

$$1 \to D^{\ell}_{A,\mathrm{iner.}} \to \pi_1(A^{\log}, a^{\log})/H \to G_k \to 1,$$

(respectively, the exact sequence of profinite groups

$$1 \to D^{\ell}_{A,\mathrm{b.iner.}} \to \pi_1(A, a)/H \to G_k \to 1,)$$

where H is the kernel of the natural morphism $\mathrm{Ker}(\pi_1(A^{\log}, a^{\log}) \to G_k) \to D^{\ell}_{A,\text{iner.}}$ (respectively, $\mathrm{Ker}(\pi_1(A, a) \to G_k) \to D^{\ell}_{A,\text{b.iner.}}$). Then we shall refer to the profinite group $\pi_1(A^{\log}, a^{\log})/H$ (respectively, $\pi_1(A, a)/H$) arising from the diagram as the **ℓ-tamely reduced geometrically pro-ℓ decomposition group** of A (respectively, **ℓ-tamely reduced geometrically pro-ℓ base-decomposition group** of A), and we shall write $D^{(\ell)}_{A,\ell\text{-tame}}$ (respectively, $D^{(\ell)}_{A,\text{b.}\ell\text{-tame}}$) for the ℓ-tamely reduced geometrically pro-ℓ decomposition group of A. We abbreviate the term "ℓ-tamely reduced geometrically pro-ℓ decomposition group" (respectively, "ℓ-tamely reduced geometrically pro-ℓ base-decomposition group") to "**ℓ-TR geometrically pro-ℓ decomposition group**" (respectively, "**ℓ-TR geometrically pro-ℓ base-decomposition group**").

(iv) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$. Then we shall refer to the maximal pro-ℓ quotient of the kernel of the morphism $\pi_1(A^{\log}, a^{\log}) \to \pi_1(\mathrm{Spec}\,\mathcal{O}_K^{\log}, \overline{s}^{\log})$ (respectively, $\pi_1(A, a) \to G_k$) as the **geometric pro-ℓ decomposition group** of A (respectively, **geometric pro-ℓ base-decomposition group** of A), where $\overline{s}^{\log}$ is a log geometric point which is the image of $a^{\log}$ in $\mathrm{Spec}\,\mathcal{O}_K^{\log}$, and we shall write $D^{\ell}_{A,\text{geom.}}$ (respectively, Δ^{ℓ}_A) for the geometric pro-ℓ decomposition group of A (respectively, the geometric pro-ℓ base-decomposition group of A).

(v) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$. Then we shall refer to the maximal pro-ℓ quotient of the kernel of the morphism $\pi_1(A^{\log}, a^{\log}) \to G_k$ as the **inertia pro-ℓ decomposition group** of A, and we shall write $D^{\ell}_{A,\text{iner.}}$ for the inertia pro-ℓ decomposition group of A.

(vi) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$. Then we obtain the exact sequence of profinite groups

$$1 \to D^{\ell}_{A,\text{iner.}} \to \pi_1(A^{\log}, a^{\log})/H \to G_k \to 1,$$

where H is the kernel of the natural morphism $\mathrm{Ker}(\pi_1(A^{\log}, a^{\log}) \to G_k) \to D^{\ell}_{A,\text{iner.}}$. We shall refer to the profinite group $\pi_1(A^{\log}, a^{\log})/H$ as the **ℓ-tamely reduced geometrically pro-ℓ decomposition group** of A, and we shall write $D^{(\ell)}_{A,\ell\text{-tame}}$ for the ℓ-tamely reduced geometrically pro-ℓ decomposition group of A. We abbreviate the term "ℓ-tamely reduced geometrically pro-ℓ decomposition group" to "**ℓ-TR geometrically pro-ℓ decomposition group**".

(vii) We shall refer to the maximal pro-ℓ quotient of the kernel of the morphism $\pi_1(A^{\log}, a^{\log}) \to \pi_1(A, a)$ as the **pro-ℓ inertia group** of A, and we shall write I^{ℓ}_A for the pro-ℓ inertia group of A.

(viii) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$. Then we obtain the natural morphism of profinite groups $I^{\ell}_A \to (I_k/P_k)^{\ell}$. We shall refer to the kernel of the morphism as the **geometric pro-ℓ inertia group** of A, and we shall write $I^{\ell}_{A,\text{geom.}}$ for the geometric pro-ℓ inertia group of A.

Proposition 1.32 (Decomposition groups and inertia groups associated to strata, certain exact sequences)**.** Let K be a complete discrete valuation field, and let $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ be a hyperbolic polycurve over K. Let ℓ be a prime number which is invertible in $\mathcal{O}_K$. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization, and that X is inertia pro-ℓ-exact and graphically inert. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n. Let A be a stratum of $\mathfrak{A}$, let $\overline{A}$ be the image of A in $\mathfrak{X}_{n-1}$, and let η_A be the generic point of A. Let a be a geometric point of A, and let $\overline{a}$ be the image of a in $\overline{A}$. Let $a^{\log}$ be a log geometric point of $A^{\log}$ lying over a geometric point a of A. Write $\overline{a}^{\log}$ for the image of $a^{\log}$ in $\overline{A}^{\log}$. Then the following assertions hold.

(i) The (natural) morphisms of profinite groups

$$D^{\ell}_{A,\mathrm{geom.}} \to \Delta^{\ell}_{X},$$

$$D^{\ell}_{A,\mathrm{iner.}} \to \Pi^{\ell}_{X,\mathrm{iner.}},$$

$$D^{(\ell)}_{A,\ell\text{-tame}} \to \Pi^{(\ell)}_{X,\ell\text{-tame}}$$

are injective.

(ii) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{G}$. Now we consider the (natural) commutative diagram of profinite groups

$$\begin{array}{ccccccccc} 1 & \longrightarrow & I^{\ell}_{A} & \longrightarrow & D^{\ell}_{A,\mathrm{geom.}} & \longrightarrow & \Delta^{\ell}_{A} & \longrightarrow & 1 \\ & & \downarrow \| & & \downarrow \cap & & \downarrow \cap & & \\ 1 & \longrightarrow & I^{\ell}_{A} & \longrightarrow & D^{\ell}_{A,\mathrm{iner.}} & \longrightarrow & D^{\ell}_{A,\mathrm{b.iner.}} & \longrightarrow & 1 \\ & & \downarrow \| & & \downarrow \cap & & \downarrow \cap & & \\ 1 & \longrightarrow & I^{\ell}_{A} & \longrightarrow & D^{(\ell)}_{A,\ell\text{-tame}} & \to & D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}} & \longrightarrow & 1. \end{array}$$

Then all the horizontal sequences in the diagram are exact.

(iii) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$. Now we consider the (natural) commutative diagram of profinite groups

$$\begin{array}{ccccccccc} 1 & \longrightarrow & I^{\ell}_{A,\mathrm{geom.}} & \longrightarrow & D^{\ell}_{A,\mathrm{geom.}} & \longrightarrow & \Delta^{\ell}_{A} & \longrightarrow & 1 \\ & & \downarrow \cap & & \downarrow \cap & & \downarrow \| & & \\ 1 & \longrightarrow & I^{\ell}_{A} & \longrightarrow & D^{\ell}_{A,\mathrm{iner.}} & \longrightarrow & \Delta^{\ell}_{A} & \longrightarrow & 1 \\ & & \downarrow \| & & \downarrow \cap & & \downarrow \cap & & \\ 1 & \longrightarrow & I^{\ell}_{A} & \longrightarrow & D^{(\ell)}_{A,\ell\text{-tame}} & \longrightarrow & \Pi^{(\ell)}_{A} & \longrightarrow & 1, \end{array}$$

where $\Pi_A^{(\ell)}$ is the geometrically pro-ℓ fundamental group of A. Then all the horizontal sequences in the diagram are exact.

(iv) Assume that $\mathfrak{sgn}(\eta_A)(n) = \mathrm{V}$ (respectively, $\mathfrak{sgn}(\eta_A)(n) = \mathrm{C}$; $\mathfrak{sgn}(\eta_A)(n) = \mathrm{N}$). Let $A^{\log}_{\mathrm{fib.}}$ be the log geometric fiber $A^{\log} \times_{\overline{A}^{\log}} \overline{a}^{\log}$. Then the natural sequence of profinite groups

$$1 \to \pi_1(A^{\log}_{\mathrm{fib.}}, a^{\log})^{\ell} \to D^{\ell}_{A,\mathrm{geom.}} \to D^{\ell}_{\overline{A},\mathrm{geom.}} \to 1,$$

is exact.

(v) In the notation of (iv), write $\Delta^{\ell}_{X_n/X_{n-1}}$ for the kernel of the natural morphism $\Delta^{\ell}_{X_n} \to \Delta^{\ell}_{X_{n-1}}$. By the assertion (iv), we can obtain the natural morphism of profinite groups $\pi_1(A^{\log}_{\mathrm{fib.}}, a^{\log})^{\ell} \to \Delta^{\ell}_{X_n/X_{n-1}}$. Then the morphism $\pi_1(A^{\log}_{\mathrm{fib.}}, a^{\log})^{\ell} \to \Delta^{\ell}_{X_n/X_{n-1}}$ is injective. Moreover, the image of the morphism $\pi_1(A^{\log}_{\mathrm{fib.}}, a^{\log})^{\ell} \to \Delta^{\ell}_{X_n/X_{n-1}}$ is a verticial subgroup (respectively, a cuspidal subgroup; a nodal subgroup) of $\Delta^{\ell}_{X_n/X_{n-1}}$, where we identify $\Delta^{\ell}_{X_n/X_{n-1}}$ with the [log geometric] pro-ℓ fundamental group of the log stable curve $\mathfrak{X}^{\log}_n \times_{\mathfrak{X}^{\log}_{n-1}} \overline{a}^{\log}$ over the log geometric point $\overline{a}^{\log}$.

(vi) Let r be the log-rank of $A^{\log}$. The pro-ℓ inertia group I^{ℓ}_A of A is a free $\mathbb{Z}_{\ell}$-module of rank r. Moreover, with respect to the natural action of $\pi_1(A, a)$ on I^{ℓ}_A, the pro-ℓ inertia group I^{ℓ}_A is isomorphic to the r-fold direct product of the pro-ℓ cyclotomes $\mathbb{Z}_{\ell}(1)$.

(vii) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$. Then the geometric pro-ℓ inertia group $I^{\ell}_{A,\mathrm{geom.}}$ is a free $\mathbb{Z}_{\ell}$-module of rank $r-1$, where r is the log-rank of $A^{\log}$.

Proof. We verify assertions (i), (ii), (iii), (iv), and (v) simultaneously by induction on n. The case where $n = 0$ is trivial. Assume that $n > 0$ and that the assertions hold for smaller values of n.

We first observe, in light of the definition of the stratification, that the morphism $A^{\log} \to \overline{A}^{\log}$ is a hyperbolic curve over $\overline{A}^{\log}$ if $\mathfrak{sgn}(\eta_A)(n) = \mathrm{V}$. Moreover, in that case, there is a compactification $A^{\log}_{\mathrm{cpt}}$ of $A^{\log}$ as a hyperbolic curve which is obtained as an irreducible component of $\mathfrak{X}^{\log}_n \times_{\mathfrak{X}^{\log}_{n-1}} \overline{A}^{\log}$. Now, we can easily verify that the right exact portion of the assertion (iv) of the case $\mathfrak{sgn}(\eta_A)(n) = \mathrm{V}$ (respectively, $\mathfrak{sgn}(\eta_A)(n) = \mathrm{C}$; $\mathfrak{sgn}(\eta_A)(n) = \mathrm{N}$) follows from [Hoshi1, Theorem 2] (respectively, Theorem 1.22; multiple applications of Theorem 1.22). Now, the assertion (v) follows immediately from the various definitions involved.

Next, consider the commutative diagram of profinite groups

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \pi_1(A^{\log}_{\mathrm{fib.}}, a^{\log})^{\ell} & \longrightarrow & D^{\ell}_{A,\mathrm{geom.}} & \longrightarrow & D^{\ell}_{\overline{A},\mathrm{geom.}} & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \mathrm{inj.}} & & \\
1 & \longrightarrow & \Delta^{\ell}_{X_n/X_{n-1}} & \longrightarrow & \Delta^{\ell}_{X_n} & \longrightarrow & \Delta^{\ell}_{X_{n-1}} & \longrightarrow & 1.
\end{array}$$

Here, the exactness of the top row, i.e., the content of assertion (iv) (respectively, the injectivity of the middle vertical arrow) follows immediately from the assertion (v) (respectively, the assertion (v)). Assertion (i) follows immediately from the injectivity of the morphism $D^{\ell}_{A,\mathrm{geom.}} \to \Delta^{\ell}_{X}$.

To verify the assertion (ii), it suffices to verify the exactness of the top row of the commutative diagram. However, the right exactness of the row follows immediately from Theorem 1.22. Therefore, it suffices to verify the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{A,\mathrm{geom.}}$.

For the case $\mathfrak{sgn}(\eta_A)(n) = \mathrm{V}$, the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{A,\mathrm{geom.}}$ follows immediately from the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{\underline{A},\mathrm{geom.}}$ (which is a consequence of the assertion (ii) for smaller values of n). For the case $\mathfrak{sgn}(\eta_A)(n) = \mathrm{C}$ or N, since the profinite group I^{ℓ}_{A} can be naturally identified with the pro-ℓ fundamental group of the log geometric fiber $A^{\log}_{\mathrm{fib.}}$, the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{A,\mathrm{geom.}}$ follows immediately from the assertion (iv).

To verify the assertion (iii), it suffices to verify the exactness of the middle row of the commutative diagram. However, the right exactness of the row follows immediately from Theorem 1.22. Therefore, it suffices to verify the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{A,\mathrm{iner.}}$.

For the case $\mathfrak{sgn}(\eta_A)(n) = \mathrm{V}$, the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{A,\mathrm{iner.}}$ follows immediately from the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{\underline{A},\mathrm{iner.}}$ (which is a consequence of the assertion (iii) for smaller values of n). For the case $\mathfrak{sgn}(\eta_A)(n) = \mathrm{C}$ or N, since the profinite group I^{ℓ}_{A} can be naturally identified with the pro-ℓ fundamental group of the log geometric fiber $A^{\log}_{\mathrm{fib.}}$, the injectivity of the morphism $I^{\ell}_{A} \to D^{\ell}_{A,\mathrm{geom.}}$ follows immediately from the assertion (iv) and the surjectivity of the morphism $D^{\ell}_{A,\mathrm{iner.}} \to (I_k/P_k)^{\ell}$ (which is a consequence of the geometric connectedness of the morphism $A \to \operatorname{Spec} k$ and the surjective portion of Theorem 1.22).

Assertion (vi) and (vii) follow immediately from the various computations of the various log structures involved (cf. Lemma 1.19). This completes the proof of the proposition. $\square$

Definition 1.33. Let K be a field. Let ℓ be a prime number. We shall say that K is **ℓ-cyclotomically full** if the ℓ-adic cyclotomic character $\chi_{K,\ell} \colon G_K \to \mathbb{Z}_{\ell}^{\times}$ of K has an open image.

Remark 1.34. Let K be a complete discrete valuation field, and let k be the residue field of K. Let ℓ be a prime number which is invertible in $\mathcal{O}_K$. Then K is ℓ-cyclotomically full if and only if k is ℓ-cyclotomically full. In particular, it follows from the fact that the ℓ-adic cyclotomic character $\chi_{K,\ell} \colon G_K \to \mathbb{Z}_{\ell}^{\times}$ of K factors through the specialization morphism $G_K \to G_k$, which is (obviously) surjective.

Proposition 1.35 (Computation of the centers of various decomposition groups)**.** Let K be a complete discrete valuation field, and let $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ be a hyperbolic polycurve over K. Let ℓ be a prime number which is invertible in $\mathcal{O}_K$. Assume that X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ with respect to its parametrization, and that X is inertia pro-ℓ-exact and graphically inert. Let $\mathfrak{A}$ be the stratification of $\mathfrak{X}_n$ obtained by applying the construction in Lemma 1.7 to the hyperbolic polycurve X_n. Let A be a stratum of $\mathfrak{A}$, let $\overline{A}$ be the image of A in $\mathfrak{X}_{n-1}$, and let η_A be the generic point of A. We identify I_A^ℓ with its images in $D^\ell_{A,\mathrm{geom.}}$, $D^\ell_{A,\mathrm{iner.}}$, and $D^{(\ell)}_{A,\ell\text{-tame}}$, as in Proposition 1.32. Then the following assertions hold.

(i) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{G}$. Then the equality $C(D^\ell_{A,\mathrm{geom.}}) = I^\ell_A$ holds.

(ii) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$. Then the equality $C(D^\ell_{A,\mathrm{iner.}}) = I^\ell_{A,\mathrm{geom.}}$ holds.

(iii) The inclusion $I^\ell_A \subset C(D^\ell_{A,\mathrm{iner.}})$ holds.

(iv) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{G}$, and that $D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}}$ is slim. Then the inequality $C(D^{(\ell)}_{A,\ell\text{-tame}}) \subset I^\ell_A$ holds. Moreover, if k is ℓ-cyclotomically full, then $D^{(\ell)}_{A,\ell\text{-tame}}$ is slim.

(v) Assume that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$, and that $\Pi^{(\ell)}_A$ is slim. Then the inequality $C(D^{(\ell)}_{A,\ell\text{-tame}}) \subset I^\ell_A$ holds. Moreover, if k is ℓ-cyclotomically full, then $\Pi^{(\ell)}_A$ is slim.

Proof. First, assuming that $\mathfrak{sgn}(\eta_A)(0) = \mathrm{G}$ (respectively, $\mathfrak{sgn}(\eta_A)(0) = \mathrm{S}$), in light of the definition of the stratification, we observe that A has a natural structure of a hyperbolic polycurve over K (respectively, a hyperbolic polycurve over k). Moreover, by Proposition 1.32, (iv), we can easily verify that the polycurve A over K (respectively, the polycurve A over k) is geometrically pro-ℓ-exact. Therefore, by Lemma 1.25, Δ^ℓ_A is slim.

Assertion (iii) follows immediately from Proposition 1.32, (vi). To verify the assertion (i) (respectively, the assertion (ii)), therefore, it suffices to verify the inclusion $C(D^\ell_{A,\mathrm{geom.}}) \subset I^\ell_A$ (respectively, the inclusion $C(D^\ell_{A,\mathrm{iner.}}) \subset I^\ell_{A,\mathrm{geom.}}$). However, this follows immediately from the slimness of Δ^ℓ_A and Proposition 1.32, (ii) (respectively, Proposition 1.32, (iii)).

The former portion of the assertion (iv) (respectively, the assertion (v)) follows from the slimness of $D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}}$ (respectively, $\Pi^{(\ell)}_A$) and Proposition 1.32, (ii) (respectively, Proposition 1.32, (iii)).

Finally, we consider the latter portion of the assertion (iv) (respectively, the assertion (v)). Here, in light of Proposition 1.32 (ii) (respectively, Proposition 1.32 (iii)), there is a natural action of $D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}}$ (respectively, $\Pi^{(\ell)}_A$) on the pro-ℓ inertia group I^ℓ_A of A. Moreover, it follows immediately from the various definition involved that the center of $D^{(\ell)}_{A,\ell\text{-tame}}$ (respectively, $D^\ell_{A,\mathrm{geom.}}$) coincides with the invariant subgroup $(I^\ell_A)^{D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}}}$ (respectively, $(I^\ell_A)^{\Pi^{(\ell)}_A}$). However, since the image of the ℓ-adic cyclotomic character $\chi_{k,\ell}\colon G_k \to \mathbb{Z}_\ell^\times$ of k is open and the action of $D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}}$ (respectively, $\Pi^{(\ell)}_A$) on I^ℓ_A is induced by the character $D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}} \to G_{K,\ell\text{-tame}} \to G_k \to \mathbb{Z}_\ell^\times$ (respectively, the character $\Pi^{(\ell)}_A \to G_k \to \mathbb{Z}_\ell^\times$), it follows from the easy computation that the invariant subgroup $(I^\ell_A)^{D^{(\ell)}_{A,\mathrm{b}.\ell\text{-tame}}}$ (respectively, $(I^\ell_A)^{\Pi^{(\ell)}_A}$) is trivial. This completes the proof of the proposition. $\square$

2 Dehn-twist Pointed Stable Curves type Outer Representations

In this section, first, we review the basic theory of profinite Dehn multi-twists (cf. [CbTpI, §4]). We then introduce the notion of Dehn-twist pointed stable curves type outer representations as a certain generalization of IPSC-type [cf. [NodNon, Definition 2.4, (i)]], SVA-type [cf. [NodNon, Definition 2.4, (ii)]] and SNN-type [cf. [NodNon, Definition 2.4, (iii)]] outer representations, and investigate their basic properties. Finally, we investigate the natural outer representations of semi-graph of anabelioids associated to the strata defined in the previous section, and in particular verify that the outer representations of inertia groups are of Dehn-twist pointed stable curves type.

Definition 2.1. Let Σ be a nonempty set of prime numbers, and let $\mathcal{G}$ be a semi-graph of anabelioids of pro-Σ PSC-type. Then we shall write

$$\mathrm{Dehn}(\mathcal{G}) \coloneqq \{\alpha \in \mathrm{Aut}^{|\mathrm{grph}|}(\mathcal{G}) \mid \alpha_{\mathcal{G}|_v} = \mathrm{id}_{\mathcal{G}|_v} \text{ for any } v \in v(\mathcal{G})\}$$

— where we refer to [CbTpI, Definition 2.1, (iii)], [CbTpI, Definition 2.14, (ii)], [CbTpI, Remark 2.5.1]. We shall refer to an element of $\mathrm{Dehn}(\mathcal{G})$ as a **profinite Dehn multi-twist** of $\mathcal{G}$.

Definition 2.2. Let Σ be a nonempty set of prime numbers, and let $\mathcal{G}$ be a semi-graph of anabelioids of pro-Σ PSC-type. Write $\widetilde{\mathcal{G}} \to \mathcal{G}$ for the universal [pro-]covering of $\mathcal{G}$ corresponding to $\Pi_{\mathcal{G}}$. Let $\widetilde{e} \in \mathrm{Node}(\widetilde{\mathcal{G}})$, $e \coloneqq \widetilde{e}(\mathcal{G}) \in \mathrm{Node}(\mathcal{G})$. Write $\widetilde{v_1}, \widetilde{v_2} \in \mathcal{V}(\widetilde{e})$ for the two vertices of $\widetilde{\mathcal{G}}$ adjacent to $\widetilde{e}$. Let $M_1, M_2 \coloneqq \Pi_{\widetilde{e}}$ (respectively, $N_1, N_2 \coloneqq \Pi_e$). Write Δ_M (respectively, Δ_N) be the image of the "diagonal" embedding $\Pi_{\widetilde{e}} \to M_1 \times M_2$; $x \mapsto (x, x)$ (respectively, $\Pi_e \to N_1 \times N_2$; $x \mapsto (x, x)$). We shall refer to $M_1 \times M_2/\Delta_M$ (respectively, $N_1 \times N_2/\Delta_N$) as the **signed nodal group** associated to $\widetilde{e}$ (respectively, e), and denote it by $\Pi_{\widetilde{e},\mathrm{sgn}}$ (respectively, $\Pi_{e,\mathrm{sgn}}$). Let $x \in \Pi_{\widetilde{e}}$ (respectively, $x \in \Pi_e$). We shall denote the image of $(x, 1) \in M_1 \times M_2$ (respectively, $(x, 1) \in N_1 \times N_2$) in $\Pi_{\widetilde{e},\mathrm{sgn}}$ (respectively, $\Pi_{e,\mathrm{sgn}}$) by $[x, \widetilde{v_1}]$ (respectively, $[x, v_1]$), and the image of $(1, x) \in M_1 \times M_2$ (respectively, $(1, x) \in N_1 \times N_2$) in $\Pi_{\widetilde{e},\mathrm{sgn}}$ (respectively, $\Pi_{e,\mathrm{sgn}}$) by $[x, \widetilde{v_2}]$ (respectively, $[x, v_2]$).

Remark 2.3. In the notation of Definition 2.2, the signed nodal group $\Pi_{\widetilde{e},\mathrm{sgn}}$ associated to $\widetilde{e}$ can be naturally identified with the signed nodal group $\Pi_{e,\mathrm{sgn}}$ associated to e.

Remark 2.4. By [CbTpI, Corollary 3.9, (vi)], there exists a natural isomorphism $\Pi_{e,\mathrm{sgn}} \cong \Lambda_{\mathcal{G}}$ (synchronization of cyclotomes), where we write $\Lambda_{\mathcal{G}}$ for the cyclotome associated to $\mathcal{G}$ (cf. [CbTpI, Definition 3.8, (i)]).

Definition 2.5. Let Σ be a nonempty set of prime numbers, and let $\mathcal{G}$ be a semi-graph of anabelioids of pro-Σ PSC-type.

(i) Let $\alpha \in \mathrm{Dehn}(\mathcal{G})$ be a profinite Dehn multi-twist of $\mathcal{G}$. Let $\widetilde{e} \in \mathrm{Node}(\widetilde{\mathcal{G}})$, $e \coloneqq \widetilde{e}(\mathcal{G}) \in \mathrm{Node}(\mathcal{G})$. We define an element of $\Pi_{\widetilde{e},\mathrm{sgn}}$ as follows.

- Let $\widetilde{v} \in \mathcal{V}(\widetilde{e})$. Then there exists a unique lifting $\alpha[\widetilde{v}] \in \mathrm{Aut}(\Pi_{\mathcal{G}})$ of α such that α preserves $\Pi_{\widetilde{v}}$ and induces the identity on $\Pi_{\widetilde{v}}$. Here, letting $\widetilde{w} \in \mathcal{V}(\widetilde{e})$ be the other vertex of $\widetilde{\mathcal{G}}$ adjacent to $\widetilde{e}$, α induces an inner automorphism of $\Pi_{\widetilde{w}}$. Therefore, there exists a unique element $\delta_{\alpha,\widetilde{v}} \in \Pi_{\widetilde{e}}$ such that $\alpha[\widetilde{v}]|_{\Pi_{\widetilde{w}}}$ coincides with the inner automorphism of $\Pi_{\widetilde{w}}$ arising from the conjugation action of $\delta_{\alpha,\widetilde{v}}$. We shall denote the element $[\delta_{\alpha,\widetilde{v}}, \widetilde{v}]$ of $\Pi_{\widetilde{e},\mathrm{sgn}}$ by $\delta_{\alpha,\widetilde{e}}$.

We note, in light of [CbTpI, Lemma 4.6], that this definition is well-defined. We denote the morphism $\mathrm{Dehn}(\mathcal{G}) \to \Pi_{\widetilde{e},\mathrm{sgn}}$ given by assigning $\alpha \mapsto \delta_{\alpha,\widetilde{e}}$ by $\mathfrak{D}_{\widetilde{e}}$. Moreover, the composite morphism $\mathrm{Dehn}(\mathcal{G}) \xrightarrow{\mathfrak{D}_{\widetilde{e}}} \Pi_{\widetilde{e},\mathrm{sgn}} \cong \Pi_{e,\mathrm{sgn}}$ depends only on e (cf. [CbTpI, Lemma 4.6]), and we shall denote it by $\mathfrak{D}_e$.

(ii) We shall write

$$\mathfrak{D}_{\mathcal{G},\mathrm{sgn.}} := \bigoplus_{e \in \mathrm{Node}(\mathcal{G})} \mathfrak{D}_e : \mathrm{Dehn}(\mathcal{G}) \to \bigoplus_{e \in \mathrm{Node}(\mathcal{G})} \Pi_{e,\mathrm{sgn}}.$$

Lemma 2.6 (Description of profinite Dehn multi-twists)**.** Let Σ be a nonempty set of prime numbers, and let $\mathcal{G}$ be a semi-graph of anabelioids of pro-Σ PSC-type. Then the following hold.

(i) The morphism $\mathfrak{D}_{\mathcal{G},\mathrm{sgn.}} : \mathrm{Dehn}(\mathcal{G}) \to \bigoplus_{e \in \mathrm{Node}(\mathcal{G})} \Pi_{e,\mathrm{sgn}}$ is isomorphism.

(ii) Let $\alpha \in \mathrm{Dehn}(\mathcal{G})$ be a profinite Dehn multi-twist of $\mathcal{G}$. For $e \in \mathrm{Node}(\mathcal{G})$, write $\delta_{\alpha,e} := \mathfrak{D}_e(\alpha)$. Let $\mathcal{B}(\mathcal{G})$ be the Galois category associated to $\mathcal{G}$, $\alpha^* \colon \mathcal{B}(\mathcal{G}) \to \mathcal{B}(\mathcal{G})$ the autoequivalence of $\mathcal{B}(\mathcal{G})$ induced by α. Then α^* is isomorphic to the autoequivalence $\Psi_{\{\delta_{\alpha,e}\}_{e \in \mathrm{Node}(\mathcal{G})}}$ of $\mathcal{B}(\mathcal{G})$ given as follows:

- Let $\{S_v, \phi_e\} \in \mathcal{B}(\mathcal{G})$, where $S_v \in \mathcal{B}(\mathcal{G}|_v)$ for each vertex v; for each node e, with branches b_1, b_2 abutting to vertices v_1, v_2, respectively, $\phi_e \colon \{(b_1)_*\}^* S_{v_1} \xrightarrow{\sim} \{(b_2)_*\}^* S_{v_2}$ is an isomorphism in $\mathcal{B}(\mathcal{G}|_e)$. We define $\Psi_{\{\delta_{\alpha,e}\}_{e \in \mathrm{Node}(\mathcal{G})}}(\{S_v, \phi_e\})$ as follows. For each node e, we shall write δ_{α,v_1} for the [unique] element of $\Pi_{\widetilde{e}}$ such that $[\delta_{\alpha,v_1}, \widetilde{v_1}] = \delta_{\alpha,e}$. Then we define $\Psi_{\{\delta_{\alpha,e}\}_{e \in \mathrm{Node}(\mathcal{G})}}(\{S_v, \phi_e\})$ to be the object $\{S_v, \phi'_e\}$ of $\mathcal{B}(\mathcal{G})$, where for each node e, with branches b_1, b_2 abutting to vertices v_1, v_2, respectively, ϕ'_e is the isomorphism in $\mathcal{B}(\mathcal{G}|_e)$ obtained as the composite of the following isomorphisms: we shall refer to the composite isomorphism $\phi_e \circ \mathsf{action}_{\delta_{\alpha,v_1}} : \{(b_1)_*\}^* S_{v_1} \xrightarrow{\sim} \{(b_2)_*\}^* S_{v_2}$, ϕ'_e, where $\mathsf{action}_{\delta_{\alpha,v_1}}$ is the isomorphism of $\{(b_1)_*\}^* S_{v_1}$ arising from the action of δ_{α,v_1}.

Proof. Assertion (i) follows immediately from [CbTpI, Theorem 4.8, (iv)] and Remark 2.4.

Assertion (ii) follows immediately from the various definitions involved. $\square$

Definition 2.7. Let Σ be a nonempty set of prime numbers, and let $\mathcal{G}$ be a semi-graph of anabelioids of pro-Σ PSC-type. Let J be a profinite group which acts continuously on $\mathcal{G}$ [i.e., we are given a continuous homomorphism $\phi\colon J \to \mathrm{Aut}(\mathcal{G})$].

(i) We shall write $\Pi_{\mathcal{G}}^{J}$ for the profinite group $\Pi_{\mathcal{G}}^{\mathrm{Aut}} \times_{\mathrm{Aut}(\mathcal{G})} J$ (cf. [CmbGC, Definition 2.3]).

(ii) We shall say that the representation $\phi\colon J \to \mathrm{Aut}(\mathcal{G})$, or the outer representation $J \to \mathrm{Out}(\Pi_{\mathcal{G}})$ induced by ϕ, is of **Dehn-twist pointed stable curves type** if the inequality $\phi(J) \subset \mathrm{Dehn}(\mathcal{G})$ holds. We abbreviate the term "Dehn-twist pointed stable curves type" to "**DTPSC-type**".

(iii) Let r be an integer. We shall say that the representation $\phi\colon J \to \mathrm{Aut}(\mathcal{G})$, or the outer representation $J \to \mathrm{Out}(\Pi_{\mathcal{G}})$ induced by ϕ, is of **Dehn-twist pointed stable curves type of rank** r if the representation is of DTPSC-type, and J is isomorphic to the free pro-Σ abelian group of rank r. We abbreviate the term "Dehn-twist pointed stable curves type of rank r" to "**DTPSC-type of rank** r".

(iv) Let J' be a subgroup of J. We shall refer to the representation $J' \to \mathrm{Aut}(\mathcal{G})$ obtained by restricting ϕ to J' as the **restriction** of ϕ to J'.

(v) Let $e \in \mathrm{Node}(\mathcal{G})$. Assume that ϕ is of DTPSC-type. Then we shall say that the representation ϕ is **degenerate at** e if the image of the composite morphism $J \xrightarrow{\phi} \mathrm{Dehn}(\mathcal{G}) \xrightarrow{\mathfrak{D}_e} \Pi_{e,\mathrm{sgn}}$ is not open. We shall say that ϕ is **non-degenerate at** e if ϕ is not degenerate at e. We shall say that ϕ is **non-degenerate** if ϕ is non-degenerate at e for any $e \in \mathrm{Node}(\mathcal{G})$.

(vi) Assume that ϕ is of DTPSC-type. Let D be a subset of $\mathrm{Node}(\mathcal{G})$. Assume that ϕ is degenerate at e for any $e \in D$. Write $\mathcal{G}_{\rightsquigarrow D}$ for the generization of $\mathcal{G}$ with respect to D (cf. [CbTpI, Definition 2.8]). Then, by [CbTpI, Theorem 4.8, (ii)], the representation ϕ induces a natural representation $J \to \mathrm{Dehn}(\mathcal{G}_{\rightsquigarrow D}) \subset \mathrm{Aut}(\mathcal{G}_{\rightsquigarrow D})$. We shall refer to the representation $J \to \mathrm{Aut}(\mathcal{G}_{\rightsquigarrow D})$ as the **generization** of ϕ with respect to D.

Remark 2.8. We note that every outer representation of DTPSC-type has an [unique] generization which is non-degenerate.

Remark 2.9. Assume $\#\Sigma = 1$. Then, it follows immediately from the various definitions involved that the image of the morphism $J \to \Pi_{e,\mathrm{sgn}}$ is not open if and only if the morphism $J \to \Pi_{e,\mathrm{sgn}}$ is trivial.

Remark 2.10. We note that the notion of DTPSC-type is a certain generalization of the notion of SVA-type, introduced in [NodNon, Definition 2.4, (ii)], and the property of being non-degenerate is a certain generalization of the property of being of NN-type, introduced in [NodNon, Definition 2.4, (iii)].

Definition 2.11. Let Σ be a nonempty set of prime numbers, and let $\mathcal{G}$ be a semi-graph of anabelioids of pro-Σ PSC-type. Let J be a profinite group which acts continuously on $\mathcal{G}$. Assume that the representation $\phi\colon J \to \mathrm{Aut}(\mathcal{G})$ is of DTPSC-type. In light of Definition 2.7, we have an exact sequence

$$1 \to \Pi_{\mathcal{G}} \to \Pi_{\mathcal{G}}^{J} \to J \to 1.$$

(i) Let $v \in \mathrm{Vert}(\mathcal{G})$ be a vertex and let $\Pi_v \subset \Pi_{\mathcal{G}}$ be a verticial subgroup of $\Pi_{\mathcal{G}}$ associated to v. Then we shall write

$$D_v \coloneqq N_{\Pi_{\mathcal{G}}^{J}}(\Pi_v) \subset \Pi_{\mathcal{G}}^{J} \text{ (respectively, } I_v \coloneqq Z_{\Pi_{\mathcal{G}}^{J}}(\Pi_v) \subset \Pi_{\mathcal{G}}^{J})$$

and refer to D_v (respectively, I_v) as a **decomposition subgroup** (respectively, **inertia subgroup**) of $\Pi_{\mathcal{G}}^{J}$ associated to v, or, alternatively, the decomposition subgroup (respectively, inertia subgroup) of $\Pi_{\mathcal{G}}^{J}$ associated to the verticial subgroup $\Pi_v \subset \Pi_{\mathcal{G}}$. If, moreover, the verticial subgroup Π_v is the verticial subgroup associated to an element $\widetilde{v} \in \mathrm{Vert}(\widetilde{\mathcal{G}})$, then we shall write $D_{\widetilde{v}} \coloneqq D_v$ (respectively, $I_{\widetilde{v}} \coloneqq I_v$) and refer to $D_{\widetilde{v}}$ (respectively, $I_{\widetilde{v}}$) as the decomposition subgroup (respectively, inertia subgroup) of $\Pi_{\mathcal{G}}^{J}$ associated to $\widetilde{v}$.

(ii) Let $c \in \mathrm{Cusp}(\mathcal{G})$ be a cusp and let $\Pi_c \subset \Pi_{\mathcal{G}}$ be a cuspidal subgroup of $\Pi_{\mathcal{G}}$ associated to c. Then we shall write

$$D_c \coloneqq N_{\Pi_{\mathcal{G}}^{J}}(\Pi_c) \subset \Pi_{\mathcal{G}}^{J} \text{ (respectively, } I_c \coloneqq \Pi_c \subset \Pi_{\mathcal{G}}^{J})$$

and refer to D_c (respectively, I_c) as a **decomposition subgroup** (respectively, **inertia subgroup**) of $\Pi_{\mathcal{G}}^{J}$ associated to c, or, alternatively, the decomposition subgroup (respectively, inertia subgroup) of $\Pi_{\mathcal{G}}^{J}$ associated to the cuspidal subgroup $\Pi_c \subset \Pi_{\mathcal{G}}$. If, moreover, the cuspidal subgroup Π_c is the cuspidal subgroup associated to an element $\widetilde{c} \in \mathrm{Cusp}(\widetilde{\mathcal{G}})$, then we shall write $D_{\widetilde{c}} \coloneqq D_c$ (respectively, $I_{\widetilde{c}} \coloneqq I_c$) and refer to $D_{\widetilde{c}}$ (respectively, $I_{\widetilde{c}}$) as the decomposition subgroup (respectively, inertia subgroup) of $\Pi_{\mathcal{G}}^{J}$ associated to $\widetilde{c}$.

(iii) Let $e \in \mathrm{Node}(\mathcal{G})$ be an node and let $\Pi_e \subset \Pi_{\mathcal{G}}$ be an edge subgroup of $\Pi_{\mathcal{G}}$ associated to e. Then we shall write

$$D_e \coloneqq N_{\Pi_{\mathcal{G}}^{J}}(\Pi_e) \subset \Pi_{\mathcal{G}}^{J} \text{ (respectively, } I_e \coloneqq Z_{\Pi_{\mathcal{G}}^{J}}(\Pi_e) \subset \Pi_{\mathcal{G}}^{J})$$

and refer to D_e (respectively, I_e) as a **decomposition subgroup** (respectively, **inertia subgroup**) of $\Pi_{\mathcal{G}}^{J}$ associated to e, or, alternatively, the decomposition subgroup (respectively, inertia subgroup) of $\Pi_{\mathcal{G}}^{J}$ associated to the nodal subgroup $\Pi_e \subset \Pi_{\mathcal{G}}$. If, moreover, the nodal subgroup Π_e is the nodal subgroup associated to an element $\widetilde{e} \in \mathrm{Node}(\widetilde{\mathcal{G}})$, then we shall write $D_{\widetilde{e}} \coloneqq D_e$ (respectively, $I_{\widetilde{e}} \coloneqq I_e$) and refer to $D_{\widetilde{e}}$ (respectively, $I_{\widetilde{e}}$) as the decomposition subgroup (respectively, inertia subgroup) of $\Pi_{\mathcal{G}}^{J}$ associated to $\widetilde{e}$.

Remark 2.12. We note here that while the terminology in Definition 2.11 may lack a certain internal consistency, preference was given to maintaining compatibility with the terminology used in [NodNon, Definition 2.2].

Lemma 2.13 (Basic Properties of Decomposition and Inertia Subgroups)**.** Let Σ be a nonempty set of prime numbers, and let $\mathcal{G}$ be a semi-graph of anabelioids of pro-Σ PSC-type. Let J be a profinite group which acts continuously on $\mathcal{G}$. Assume that the representation $\phi\colon J \to \mathrm{Aut}(\mathcal{G})$ is of DTPSC-type. Then the following hold.

(i) Let $\widetilde{v} \in \mathrm{Vert}(\widetilde{\mathcal{G}})$. Then the following equalities hold.

- $\Pi_{\mathcal{G}} \cap I_{\widetilde{v}} = 1$.
- $\Pi_{\mathcal{G}} \cap D_{\widetilde{v}} = \Pi_{\widetilde{v}}$.

Moreover, we have an exact sequence

$$1 \to \Pi_{\widetilde{v}} \to D_{\widetilde{v}} \to J \to 1,$$

and the morphism $I_{\widetilde{v}} \to J$ induced by the inclusion $I_{\widetilde{v}} \subset D_{\widetilde{v}}$ is an isomorphism. Thus, $D_{\widetilde{v}}$ is naturally isomorphic to the product $\Pi_{\widetilde{v}} \times I_{\widetilde{v}}$, or $\Pi_{\widetilde{v}} \times J$.

(ii) Let $\widetilde{e} \in \mathrm{Node}(\widetilde{\mathcal{G}})$, $\widetilde{v} \in \mathcal{V}(\widetilde{e})$. Then the followings hold.

- $I_{\widetilde{v}} \subset I_{\widetilde{e}}$.
- $\Pi_{\mathcal{G}} \cap I_{\widetilde{e}} = \Pi_{\widetilde{e}}$.
- $D_{\widetilde{e}} = I_{\widetilde{e}}$.

Moreover, we have an exact sequence

$$1 \to \Pi_{\widetilde{e}} \to I_{\widetilde{e}} \to J \to 1,$$

and the morphism $I_{\widetilde{v}} \times \Pi_{\widetilde{e}} \to I_{\widetilde{e}}$ induced by the inclusions $I_{\widetilde{v}} \subset I_{\widetilde{e}}$ is an isomorphism.

(iii) Let $\widetilde{c} \in \mathrm{Cusp}(\widetilde{\mathcal{G}})$, $\widetilde{v} \in \mathcal{V}(\widetilde{c})$. Then the following equality holds.

- $\Pi_{\mathcal{G}} \cap D_{\widetilde{c}} = \Pi_{\widetilde{c}}$.

Moreover, we have an exact sequence

$$1 \to \Pi_{\widetilde{c}} \to D_{\widetilde{c}} \to J \to 1,$$

and the morphism $I_{\widetilde{v}} \times \Pi_{\widetilde{c}} \to D_{\widetilde{c}}$ induced by the inclusions $I_{\widetilde{v}} \subset D_{\widetilde{c}}$ is an isomorphism.

Proof. The former portion of the assertion (i) follows immediately from the various definitions involved. To verify the latter portion of the assertion (i), it suffices to show that the morphism $I_{\widetilde{v}} \to J$ induced by the inclusion $I_{\widetilde{v}} \subset D_{\widetilde{v}}$ is an isomorphism. We note, by the equation $\Pi_{\mathcal{G}} \cap I_{\widetilde{v}} = 1$, that the morphism $I_{\widetilde{v}} \to J$ is injective. Now, we take a morphism $s\colon J \to \mathrm{Aut}(\Pi_{\mathcal{G}})$ in a following manner: for $j \in J$, we define $s(j)$ for the unique lifting of $\phi(j)$ to an element of $\mathrm{Aut}(\Pi_{\mathcal{G}})$ which preserves $\Pi_{\widetilde{v}}$ and induces the identity on $\Pi_{\widetilde{v}}$. Then, by the definitions involved, the image of s is contained in $I_{\widetilde{v}}$. Moreover, the composite morphism $J \xrightarrow{s} I_{\widetilde{v}} \to J$ is obviously the identity morphism. Thus, the morphism $I_{\widetilde{v}} \to J$ is an isomorphism, and the assertion (i) follows.

□

Lemma 2.14. Let $\widetilde{e} \in \mathrm{Node}(\widetilde{\mathcal{G}})$. Let $\widetilde{v_1}, \widetilde{v_2} \in \mathcal{V}(\widetilde{e})$ be the two vertices of $\widetilde{\mathcal{G}}$ adjacent to $\widetilde{e}$. Assume that the [outer] representation ϕ is non-degenerate at $\widetilde{e}$. Then the morphism $I_{\widetilde{v_1}} \times I_{\widetilde{v_2}} \to I_{\widetilde{e}}$ has open image.

Lemma 2.15. center とかの群論の計算、inertia の話

Proof. NodNon など □

inertia の線形代数は、かかないといけない

あとは、strata に付随する外表現 (環論との対応)

nondegenerate, とかいたが、実は SNN-type のこととあわせて修正する必要がある気がする。あと、IPSC-type にあわせたかたちのものについても、かく必要があるかもしれない - 真面目に環論をかんがえていないのでまだわからない。

3 Reconstruction of Log-full Subgroups

4 Reconstruction of Dimensions via Corank 1 Intersections

5 Reconstruction of the Graph-theoretic Structure of Proper Hyperbolic Polycurves of Dimension 2

6 Cofinality of l-admissible finite étale coverings

Definition 6.1. Let K be a field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. We shall say that X is **pro-l exact** (with respect to its parametrization) if for each $0 \leq i \leq n$, the (natural) sequence of profinite groups

$$1 \to \Delta^l_{X_n/X_i} \to \Pi^l_{X_n} \to \Pi^l_{X_i} \to 1$$

is exact. Here, Δ_{X_n/X_i} denotes the kernel of the homomorphism $\Pi_{X_n} \to \Pi_{X_i}$ induced by the morphism $X_n \to X_i$.

Definition 6.2. Let K be a valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Let l be a prime number which is different from the characteristic of the residue field of K. We shall say that X is **l-quasi-fine** (with respect to its parametrization) if X admits a compactified stable model $\mathfrak{X} = \mathfrak{X}_n \to \mathfrak{X}_{n-1} \to \cdots \to \mathfrak{X}_1 \to \mathfrak{X}_0 = \operatorname{Spec} \mathcal{O}_K$ and X is pro-l exact. We shall say that X is **l-fine** (with respect to its parametrization) if X is l-quasi-fine and for each $i = 1, \ldots, n$ and each geometric point $\overline{x}$ of X_{i-1} which lies over the closed point of $\operatorname{Spec} \mathcal{O}_K$, the geometric fiber $X_{i,\overline{x}}$ of the morphism $X_i \to X_{i-1}$ over $\overline{x}$ is singular.

Theorem 6.3. Let K be a valuation field and $X = X_n \to X_{n-1} \to \cdots \to X_1 \to X_0 = \operatorname{Spec} K$ a hyperbolic polycurve over K. Let l be a prime number which is different from the characteristic of the residue field of K. Then there exists a finite étale covering $Y \to X$ such that Y is l-fine (with respect to its parametrization induced by the morphism $Y \to X$).

Proof. proof here $\square$

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